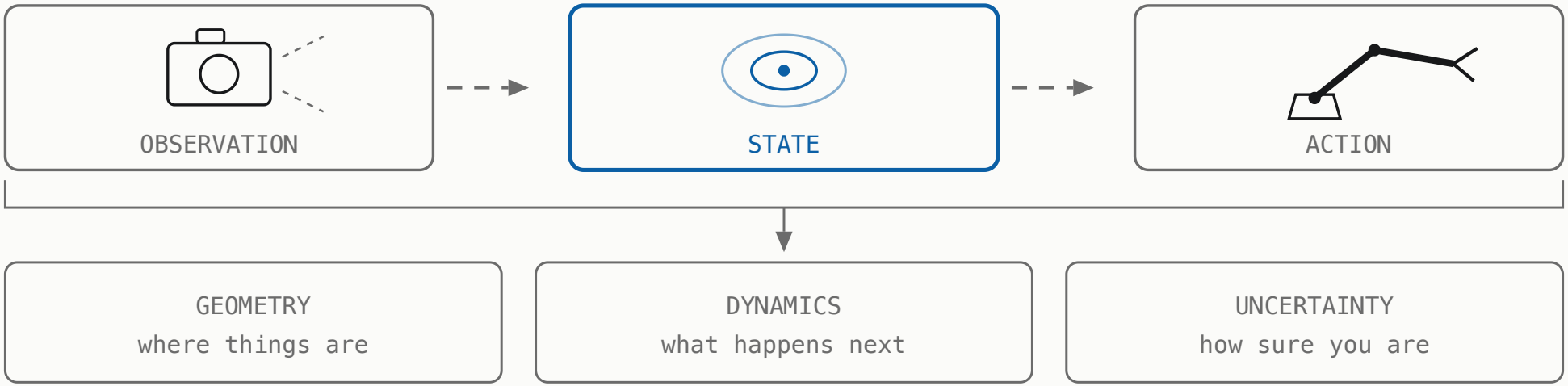
Introduction and Linear Algebra I

Spaces, Maps, Rank, Solutions

Mathematics for Robotics · ROB-GY 6013 · NYU Tandon

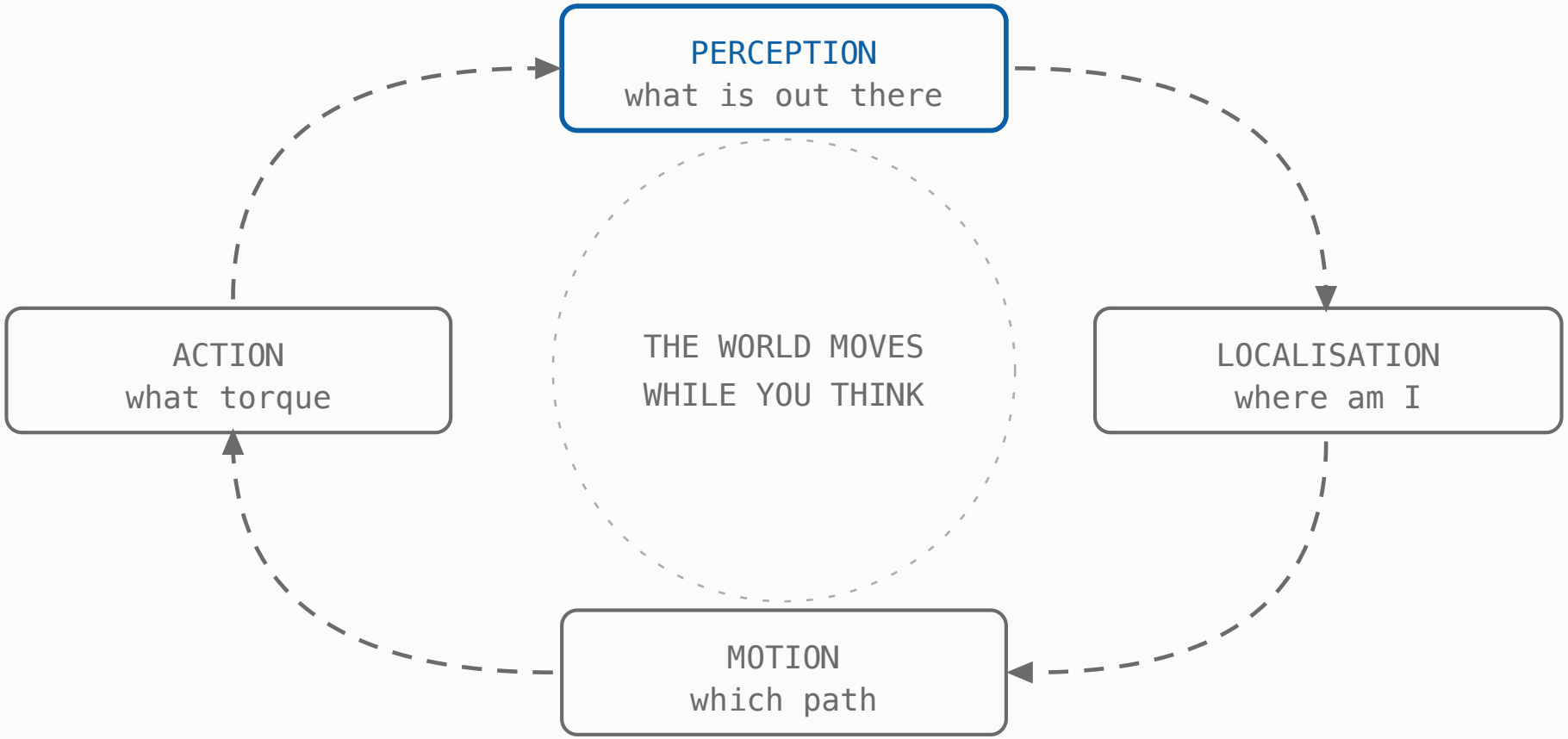
Jing Zhang · Fall 2026 · September 2

Why Mathematics for Robotics?



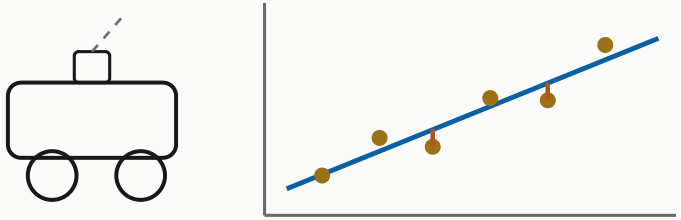
The three panels are recurring questions, not one per box above.

Why Mathematics for Robotics?



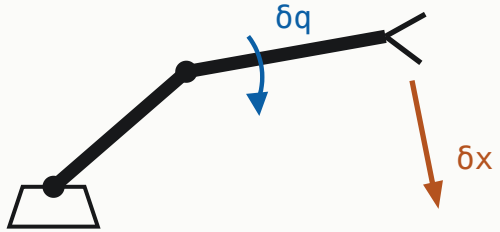
The loop closes many times a second.

Why Mathematics for Robotics?



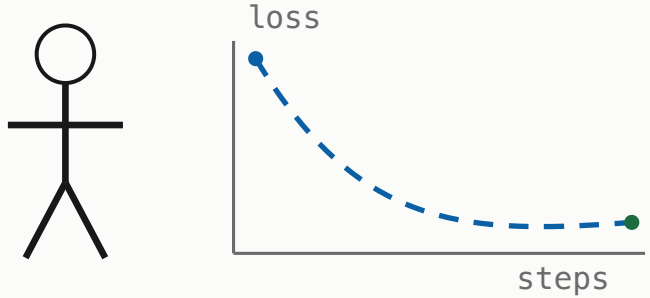
CALIBRATION

locally, $Ax \approx b$



ROBOT MOTION

$\delta x \approx J\,\delta q$



LEARNING

$\arg\min_\theta \mathcal{L}(\theta)$

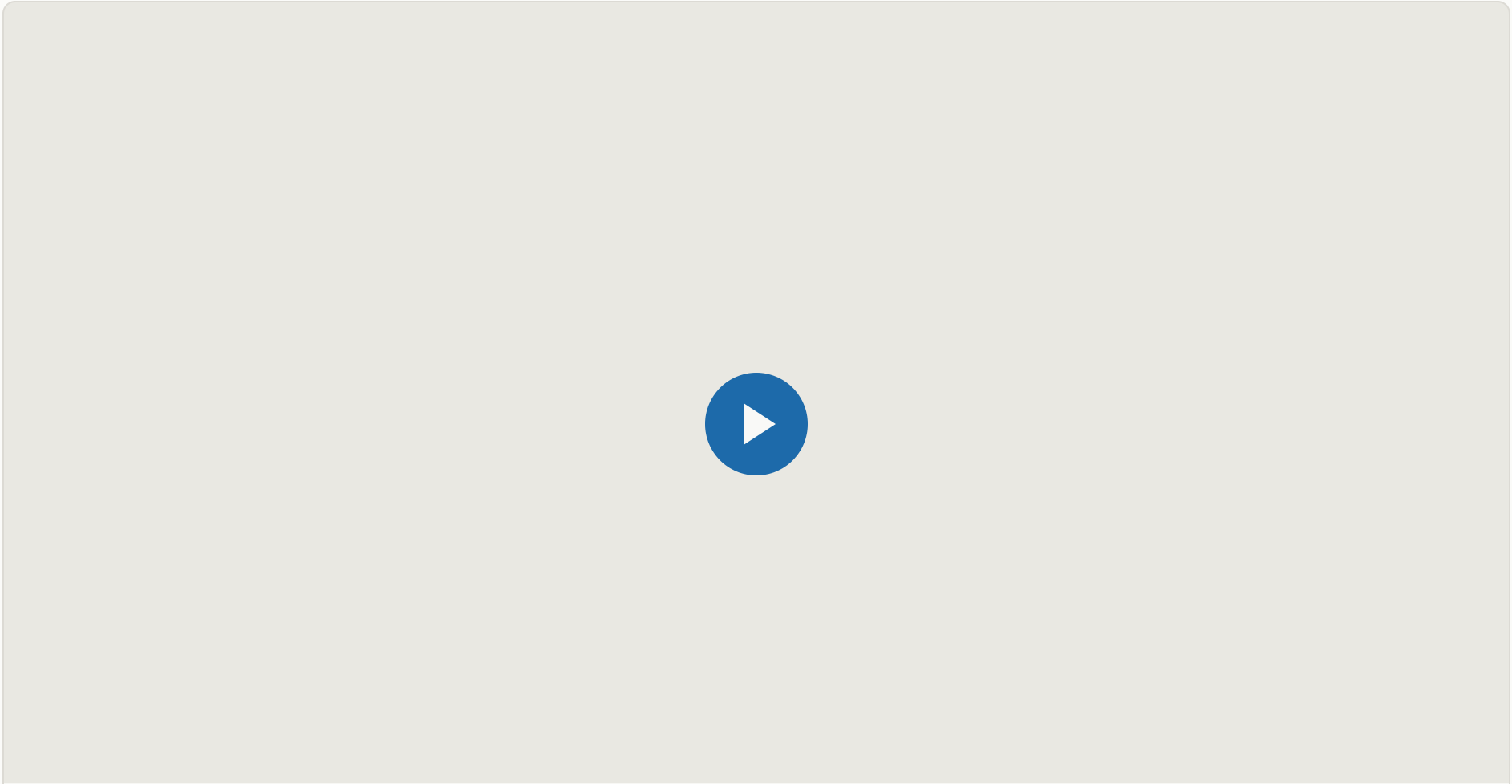
> Different surface, same mathematical structure.

Why Mathematics for Robotics?

Robotics changes quickly.

The mathematical structure changes much more slowly.

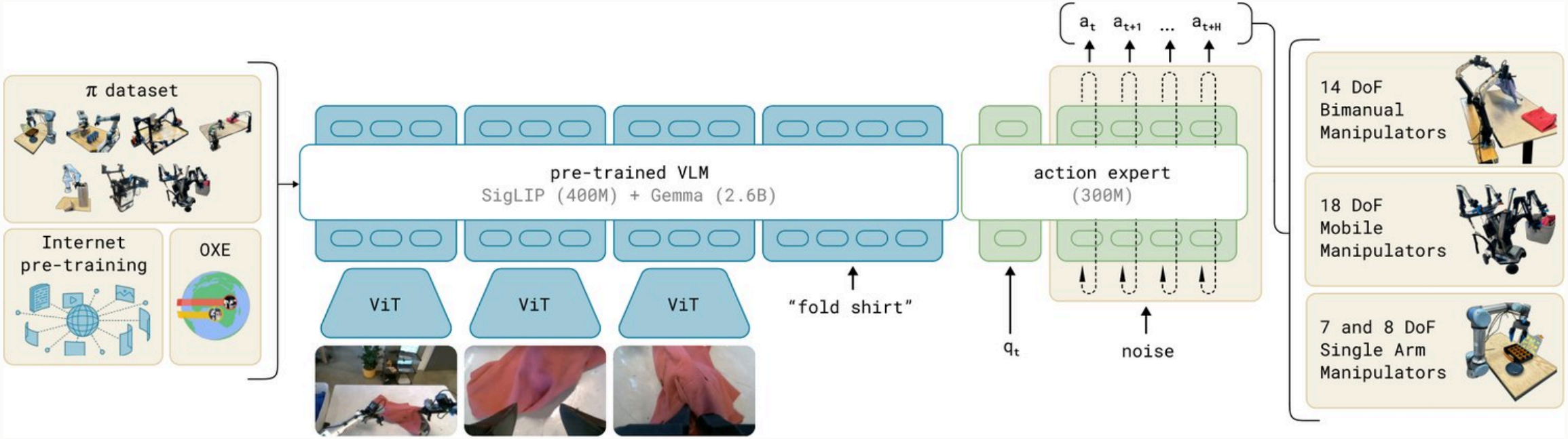
Why Mathematics in the GenAI Era?



Google DeepMind · Advanced dexterity with Gemini Robotics 2, July 2026

opens on YouTube

Why Mathematics in the GenAI Era?

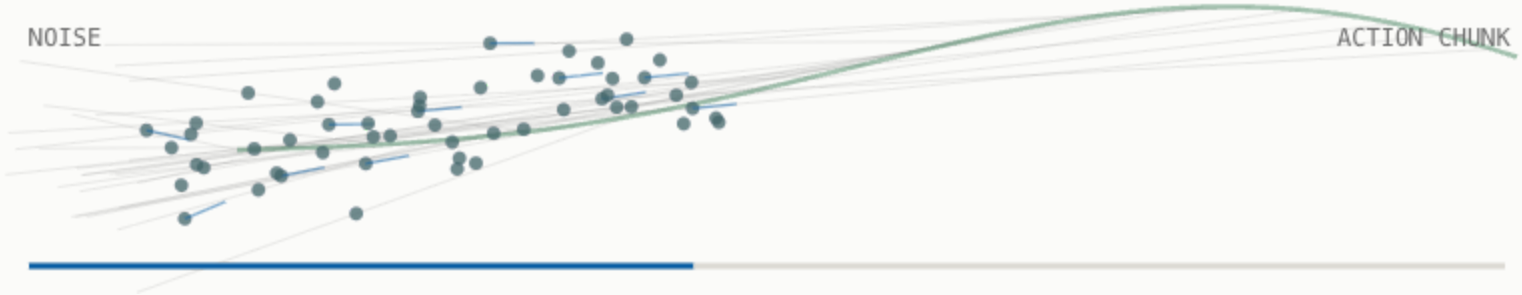


VLM · attention, masks · Transformers $\mathbf{q_t}$ · vector spaces · Linear Algebra I–IV **noise** → $\mathbf{a_t}$ · a vector field · ODE · Flow Matching

A modern VLA is not replacing mathematics. It is built from mathematics.

Black, K. et al. ***π0: A Vision-Language-Action Flow Model for General Robot Control***, Figure 3. RSS, 2025. arXiv:2410.24164.

Why Mathematics in the GenAI Era?

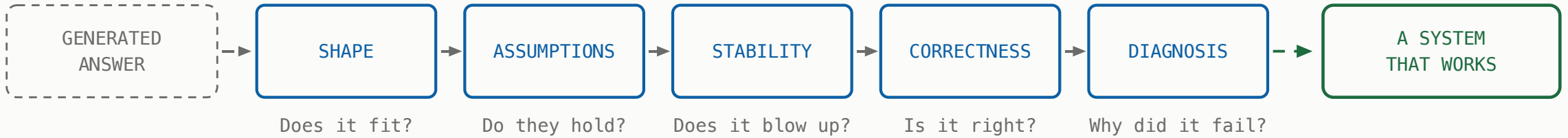


$$\boxed{\frac{dA_\tau}{d\tau} = v_\theta(A_\tau, o, \tau)} \qquad A_0 = \epsilon, \qquad \hat{a} = A_1$$

- τ flow time, 0 to 1; the field changes with it
- v_θ the trained vector field
- A_0 the start: pure noise ε
- $\hat{a} = A_1$ the action chunk, read off at $\tau = 1$

Black, K. et al. ***π0***. RSS, 2025. arXiv:2410.24164.

Why Mathematics in the GenAI Era?



> AI can generate. You still have to judge.

Why Mathematics in the GenAI Era?

WHAT AN ML INTERVIEW ASKS ABOUT		
Rank / span ●	**Null / image** ●	**Independence** ●
Inverse ●	**Singular** ●	Orthogonality
Positive semidefinite	Eigenvalues	Jacobian
Hessian	Determinant	Dot product

AND TO WRITE, BY HAND		Lectures 5 · 6 · 10
transformer end-to-end	causal, cross, self attention	flash attention
attention backward pass	MLP forward and backward	SGD training loop

> "...with AI assistance completely off."

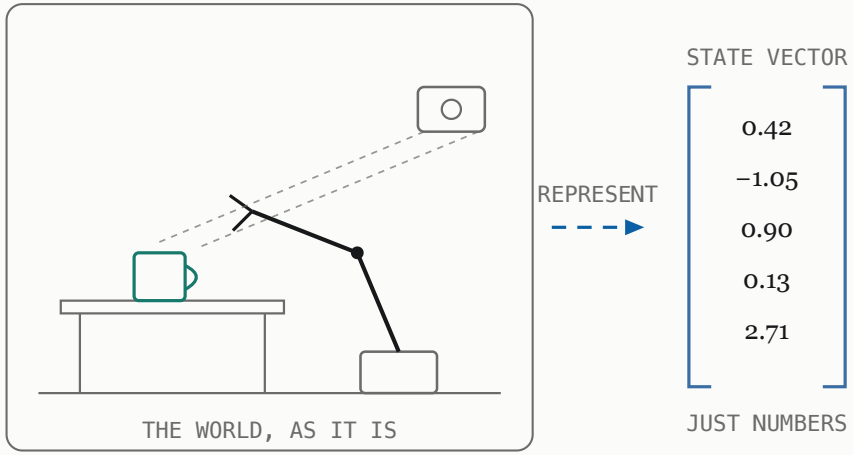
Sapora, S. ***ML Job Interviews: The Ultimate Guide***, 2026 · silviasapora.github.io/blog/ml-interviews.html
Liu, A. ***Notes on the Industry Job Search***, 2026 · alisawuffles.github.io/blog/job-search/

The Math We Keep Coming Back To

x	state	Linear Algebra I · ODE · Probability I
A	linear map	Linear Algebra I–IV
f	nonlinear function	ML Math I · ODE
J	Jacobian	Linear Algebra IV · ML Math I–II
p	probability distribution	Probability I–II · Diffusion
L	loss	ML Math II · Diffusion · Flow Matching

> Different problems, the same mathematical ideas.

The Math We Keep Coming Back To



FOUR ROBOT QUANTITIES

$q \in \mathbb{R}^7$	joint configuration
$p \in \mathbb{R}^3$	position
$I \in \mathbb{R}^{3\times H\times W}$	image
$a \in \mathbb{R}^{16\times 7}$	action chunk

Each is a vector, or an array that can be vectorized.

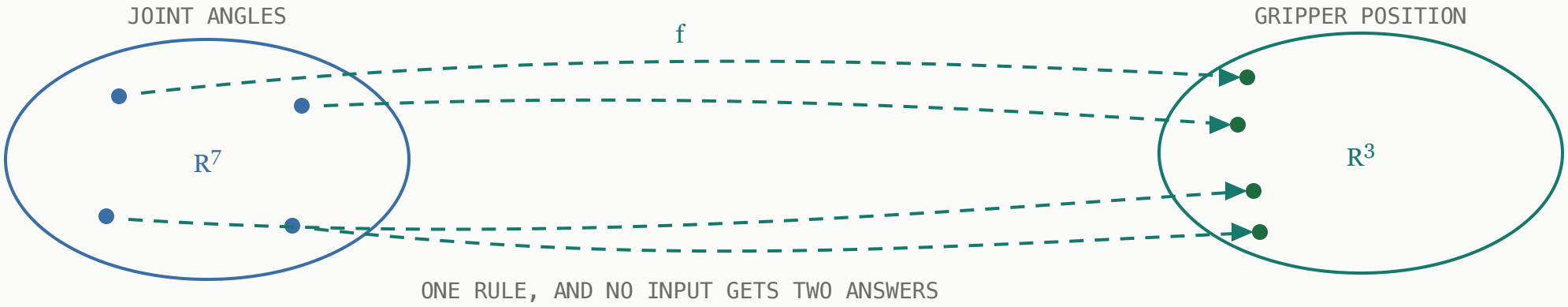
The Math We Keep Coming Back To

MAP $f : \mathbb{R}^n \to \mathbb{R}^m, \qquad u \mapsto y = f(u)$

`domain` $\mathbb{R}^n$, the allowed inputs

`codomain` $\mathbb{R}^m$, the space the outputs live in

`u ↦ y` the rule that carries one to the other



Forward kinematics: $\mathbb{R}^7 \to \mathbb{R}^3$, joint angles to gripper position.

The Math We Keep Coming Back To

LOCAL LINEARISATION

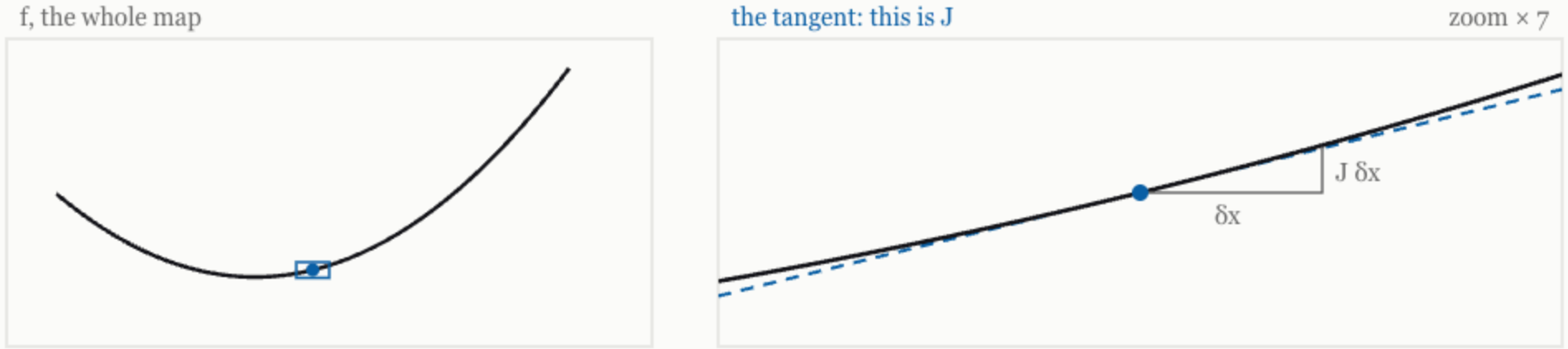
$$f(x + \delta x) \approx f(x) + J(x)\,\delta x$$

`x` the current point

`δx` a small change

`J(x)` the Jacobian, $J(x) \in \mathbb{R}^{m \times n}$

> Many nonlinear robotics problems become linear when viewed locally.

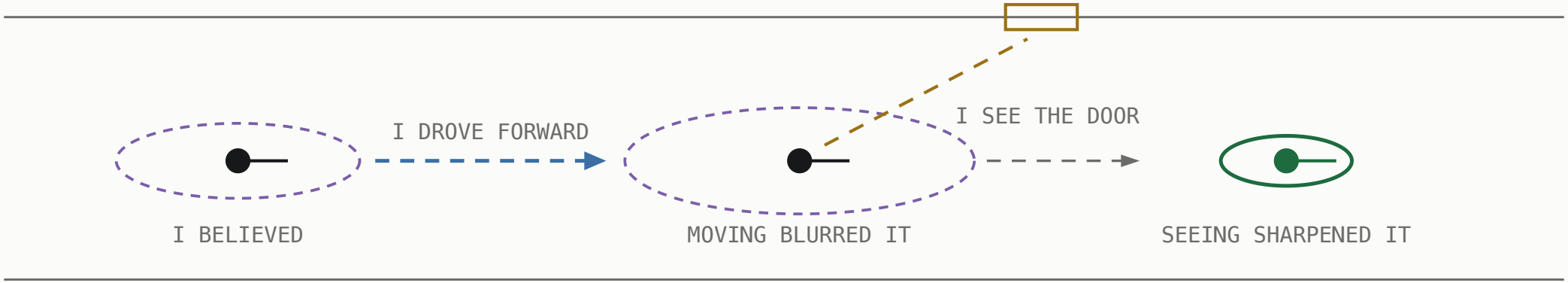


Left, the shrinking window. Right, what is inside it.

The Math We Keep Coming Back To

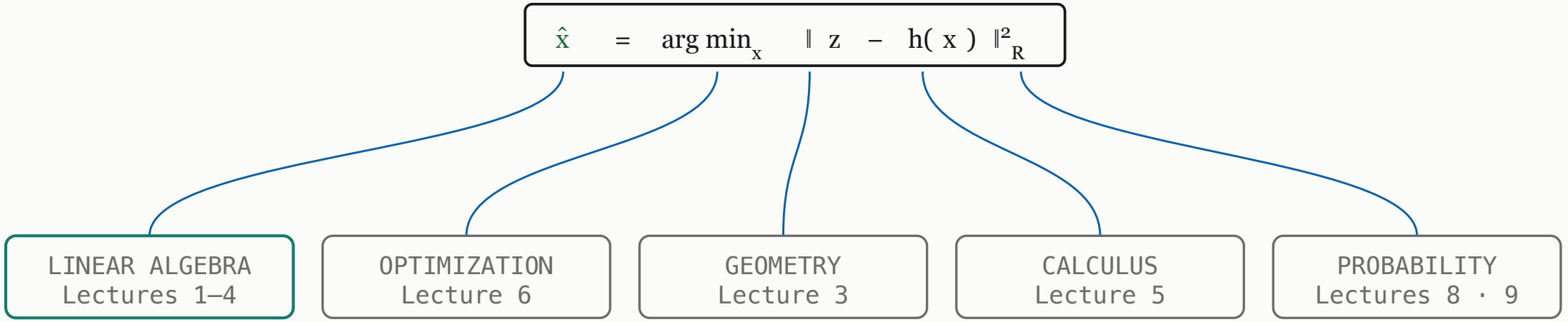
WHERE AM I? $x_t = f(x_{t-1}, u_t) + w_t, \qquad z_t = h(x_t) + v_t$

- x_t the latent state, usually not observed directly
- u_t, z_t the command, the measurement
- f, h motion and sensor models
- w_t, v_t noise



Thrun, S., Burgard, W., Fox, D. ***Probabilistic Robotics***, Ch. 7. MIT Press, 2005.

The Math We Keep Coming Back To



$\|e\|_R^2 = e^\mathsf{T} R^{-1} e$, with R the measurement covariance.

e $z - h(x)$ here, measurement minus prediction, which the course writes r; from Lecture 5 on, e is prediction minus target, so the two differ by a sign

Barfoot, T. D. ***State Estimation for Robotics***, Chapter 4. Cambridge University Press, 2017.

Course information

THE MEETING

WHEN	Wednesdays 5:00–7:30 PM
WHERE	2 MetroTech Center, Room 909
ONLINE	nyu.zoom.us/j/95222620311
UPDATES	slides and recordings on the course page
QUESTIONS	Ed Discussion, linked from the course page

THE STAFF

Jing Zhang
instructor · z.jing@nyu.edu

Shichen Zhang
teaching assistant · sz4968@nyu.edu

office hours to be announced

Everything on these four pages, plus the schedule and the reading: jingz6676.github.io/rob6013.html

Course information

GRADING

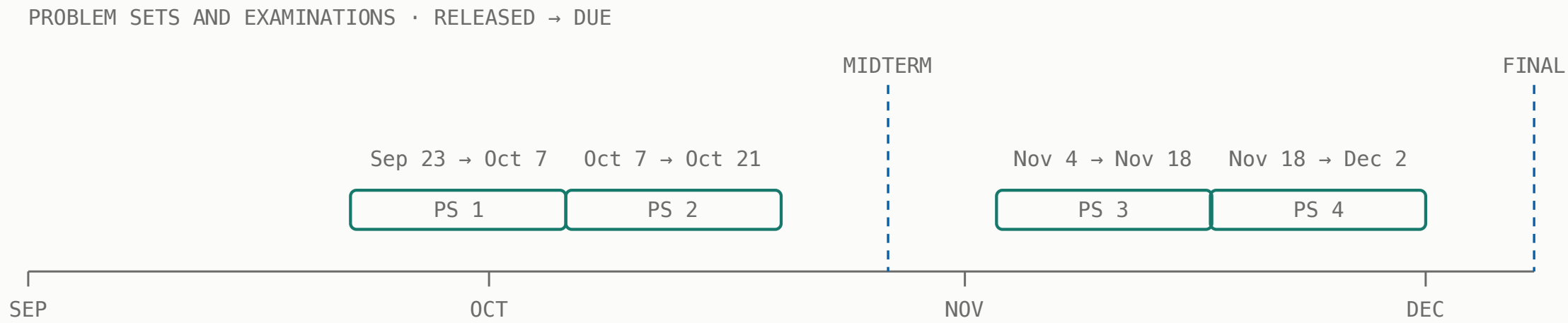


FOUR PROBLEM SETS	MIDTERM	FINAL
understanding, not recall	28 October, in person	9 December, in person

---

up to +10% on the course grade, for sets typeset in LaTeX or Markdown

Course information



> One sheet of paper, both sides, prepared by you, and a scientific calculator. No phone, laptop, tablet, or watch.

Course information

POLICIES	
LATE WORK	10% of the score per calendar day; closed once solutions are posted
COLLABORATION	discuss with anyone, then write your solution alone
USING AI	any tool to understand it, none to write your solutions
ACCOMMODATIONS	through the Moses Center, with no need to tell me why
IF IT GOES WRONG	email me and Shichen

LINEAR ALGEBRA I

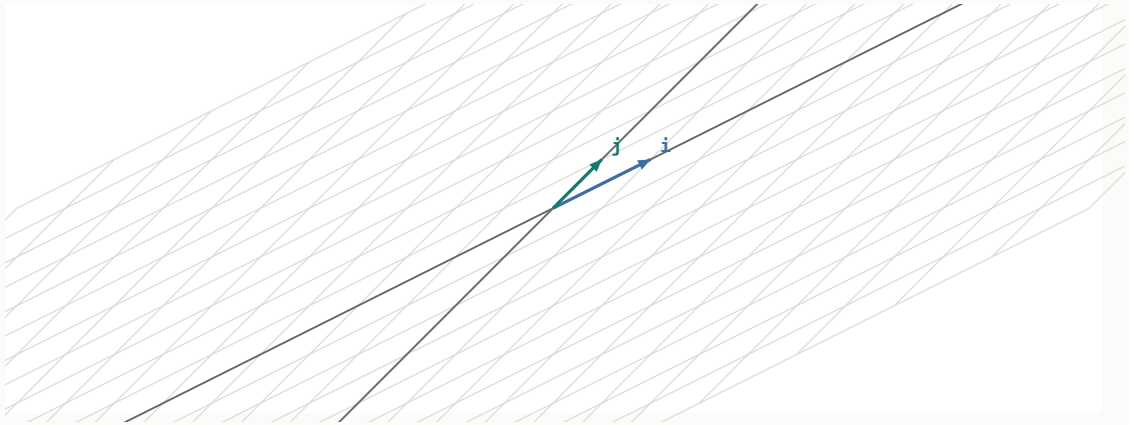
Spaces, Maps, Rank, Solutions

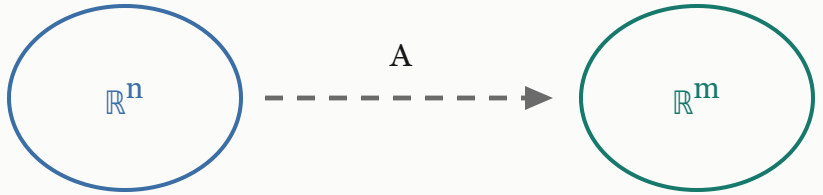


A matrix is a map

$$A \in \mathbb{R}^{m \times n}, \qquad x \in \mathbb{R}^n \longmapsto Ax \in \mathbb{R}^m$$

- `A` the map itself, m rows by n columns
- `x` the input, a column of n numbers
- `Ax` the output, a column of m numbers
- `n` input dimension: the number of columns
- `m` output dimension: the number of rows; m and n need not be equal





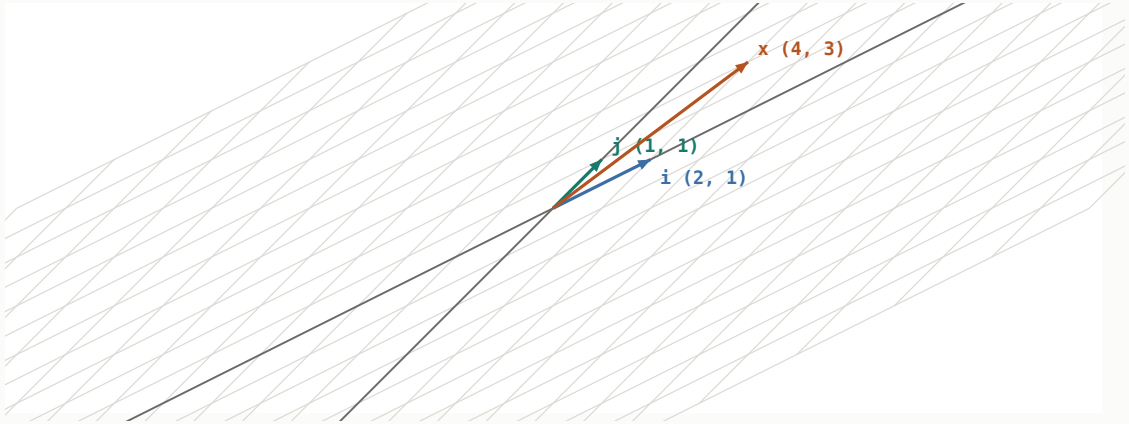
Sanderson, G. ***Essence of Linear Algebra***, ch. 3. 3Blue1Brown, 2016.

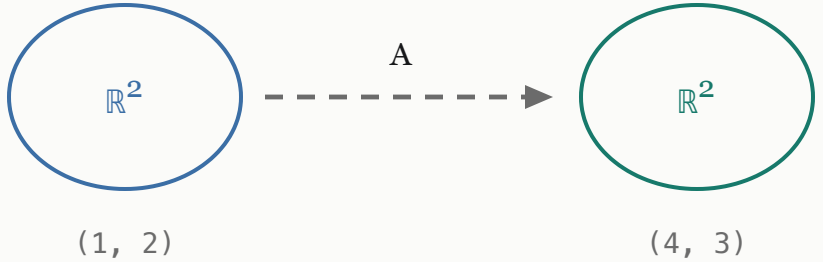
A matrix is a map

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+2 \\ 1+2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ the first column, where e_1 lands

$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ the second column, where e_2 lands





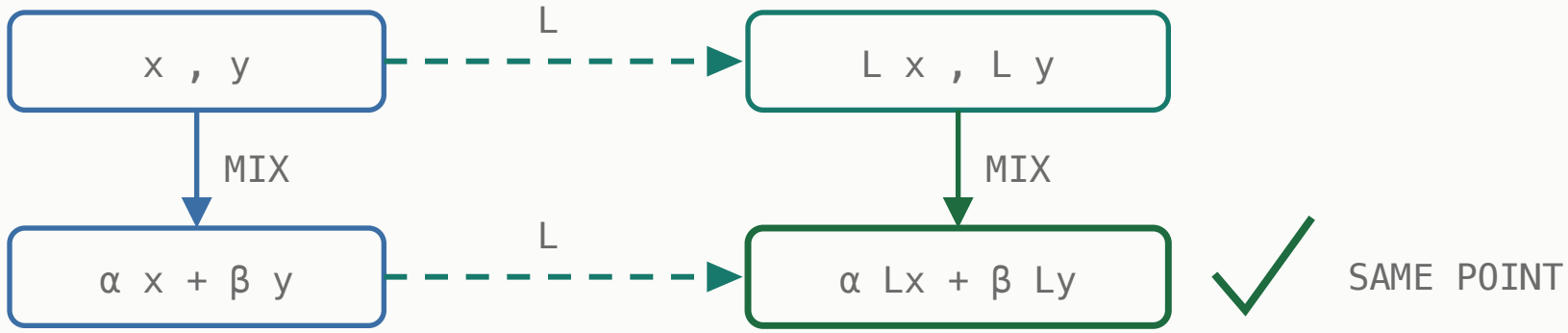
A matrix is a map

LINEAR MAP $\boxed{L(\alpha x + \beta y) = \alpha L(x) + \beta L(y)}$

additivity $L(x + y) = L(x) + L(y)$, the case $\alpha = \beta = 1$

homogeneity $L(\alpha x) = \alpha L(x)$, the case $\beta = 0$

superposition both at once, for all x, y and all $\alpha, \beta \in \mathbb{R}$

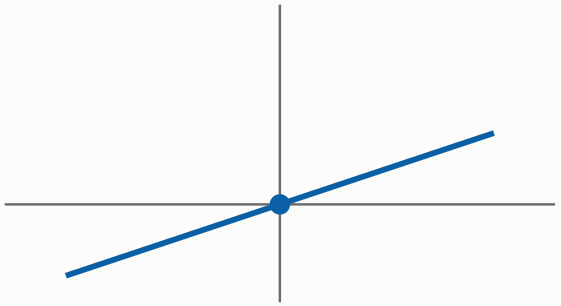


Axler, S. ***Linear Algebra Done Right***, 4th ed., Definition 3.1, p. 52. Springer, 2024. Open access.

QUESTIONS

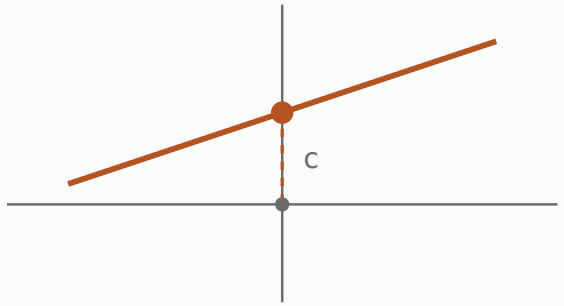
Which of these three maps is linear?

$f(x) = a\,x$



LINEAR

$f(x) = a\,x + c, \quad c \neq 0$

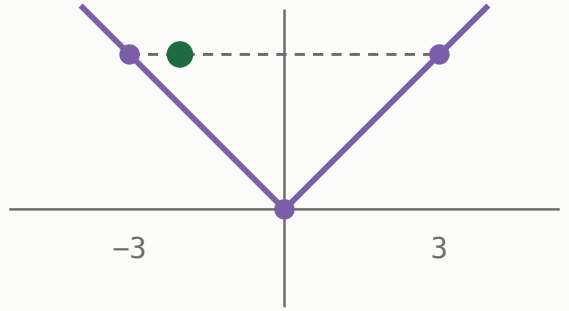


AFFINE, NOT LINEAR

$a = 1, \quad c = 2$

✗ $f(0) = 2 \neq 0$

$f(x) = |x|$



NOT LINEAR

✓ $f(0) = 0$

✗ $f(-3) = 3, \quad -f(3) = -3$

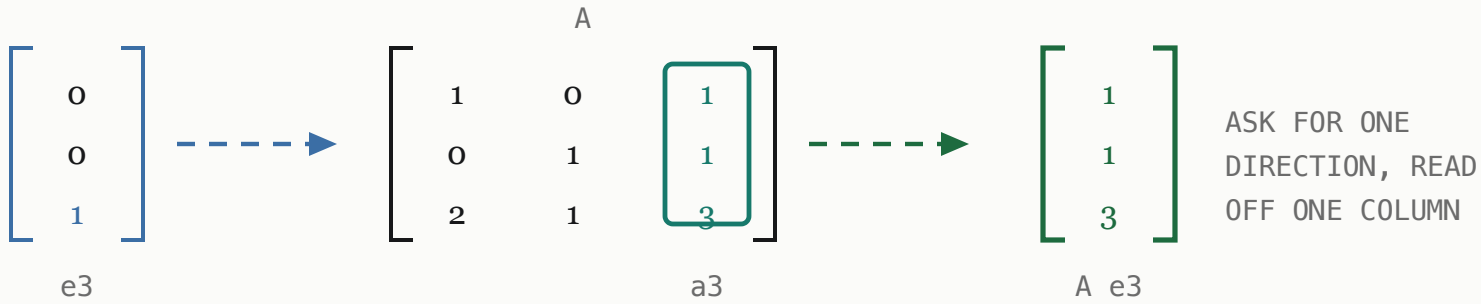
A matrix is a map

$$e_j = (0, \ldots, 1, \ldots, 0)^{\mathsf{T}}$$

the 1 sits in slot j; every other slot holds 0

COLUMN RULE $\boxed{Ae_j = a_j}$

a_j column j of A, a whole output vector



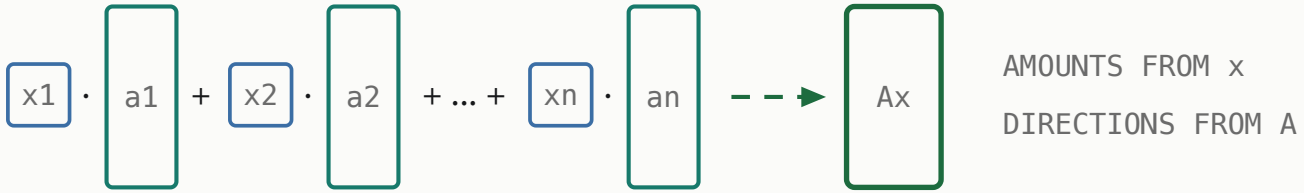
Axler, S. ***Linear Algebra Done Right***, 4th ed.: standard basis 2.27(a), p. 39; columns 3.50, p. 76. Springer, 2024.

A matrix is a map

LINEAR COMBINATION OF COLUMNS

$$Ax = \sum_{j=1}^{n} x_j \, (Ae_j) = x_1 a_1 + \cdots + x_n a_n$$

x_j	coordinate j of x, used as an amount
Ae_j	column j, from the page before
a_j	column j of A, a whole direction



AMOUNTS FROM x

DIRECTIONS FROM A

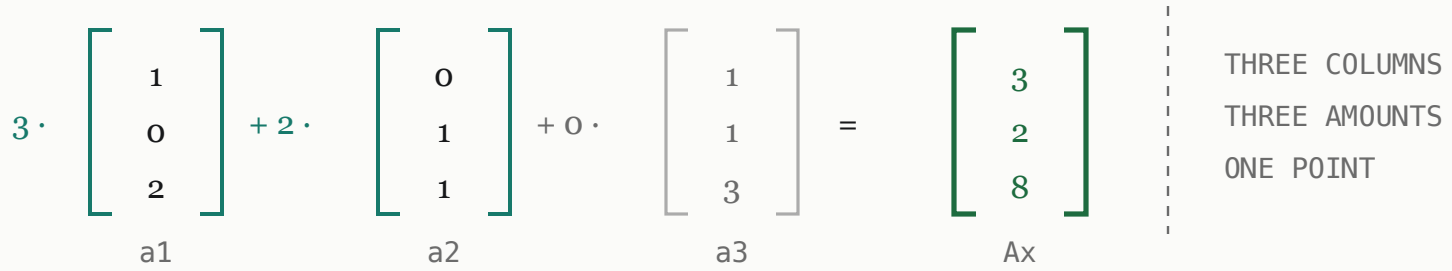
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.50 linear combination of columns, p. 76. Springer, 2024.

A matrix is a map

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = 3a_1 + 2a_2 = \begin{bmatrix} 3 \\ 2 \\ 8 \end{bmatrix}$$

`a1, a2` (1,0,2) and (0,1,1), the first two columns

`3, 2, 0` the amounts asked of the three columns



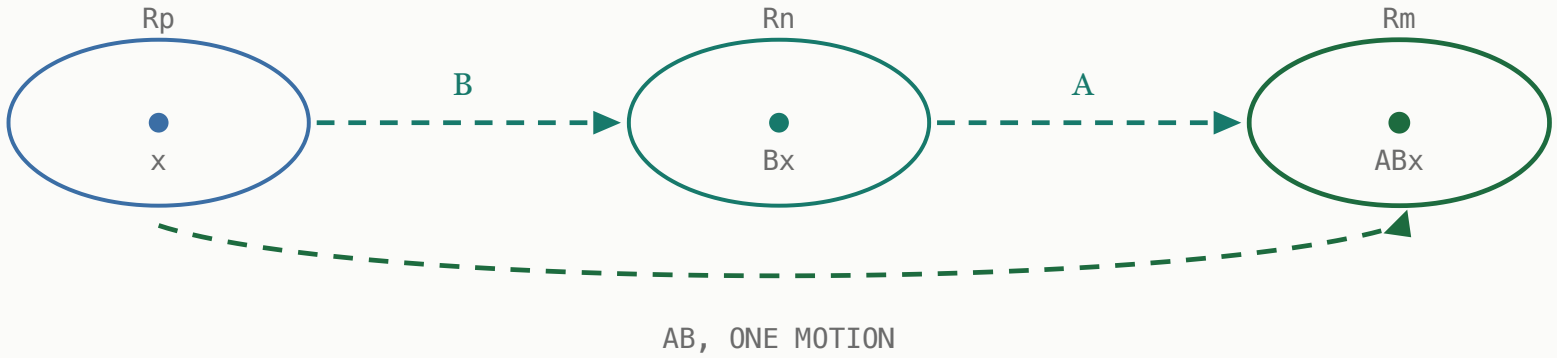
A matrix is a map

DO B FIRST, THEN A $(AB)x = A(Bx)$

shapes $B \in \mathbb{R}^{n\times p}$, $A \in \mathbb{R}^{m\times n}$

Bx where B sends x first

A(Bx) where A then sends that



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.43 matrix of product of linear maps, p. 74. Springer, 2024.

A matrix is a map

TRANSPOSE $(A^{\mathsf{T}})_{k,j} = A_{j,k}$

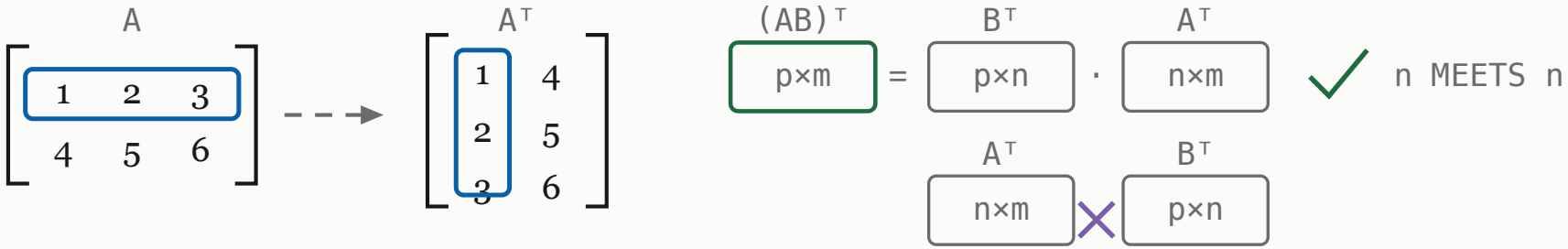
`row j` of A becomes column j of A^{T}

`m × n` becomes n × m

IT REVERSES A PRODUCT $\boxed{(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}}$

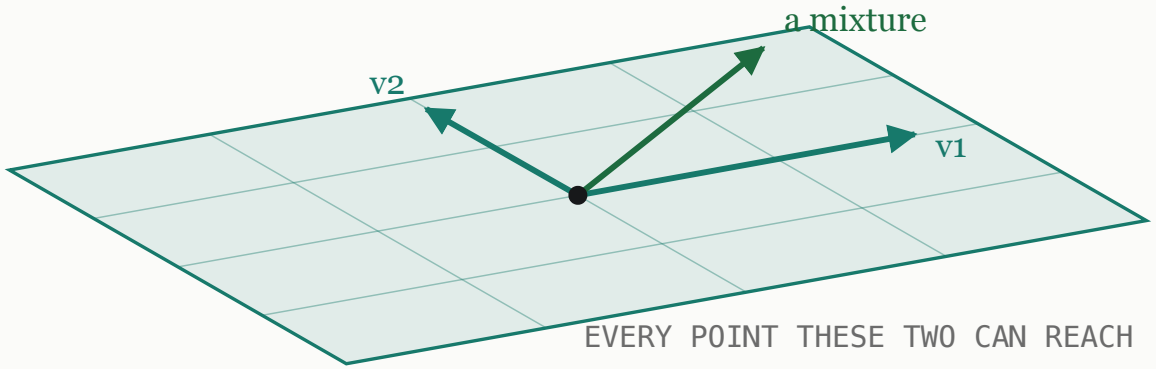
$(AB)^{\mathsf{T}}$ p × m, the shape the answer must have

$B^{\mathsf{T}}A^{\mathsf{T}}$ p × n times n × m, so n meets n



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.54 transpose, p. 77; $(AC)^t = C^tA^t$, p. 78. Springer, 2024.

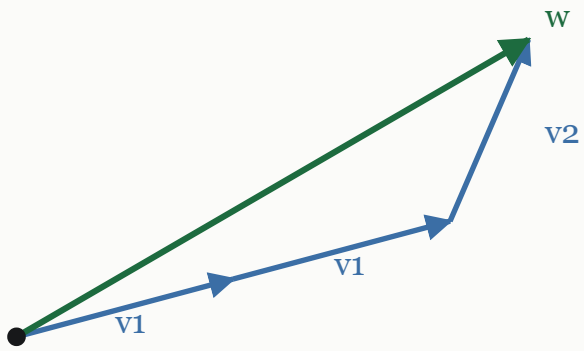
Span and column space



Span and column space

LINEAR COMBINATION $w = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k$

- v_i the directions you are handed
- c_i any reals, negative and zero included
- w where those weights land you



TWO STEPS OF v1

ONE STEP OF v2

w = 2 v1 + v2

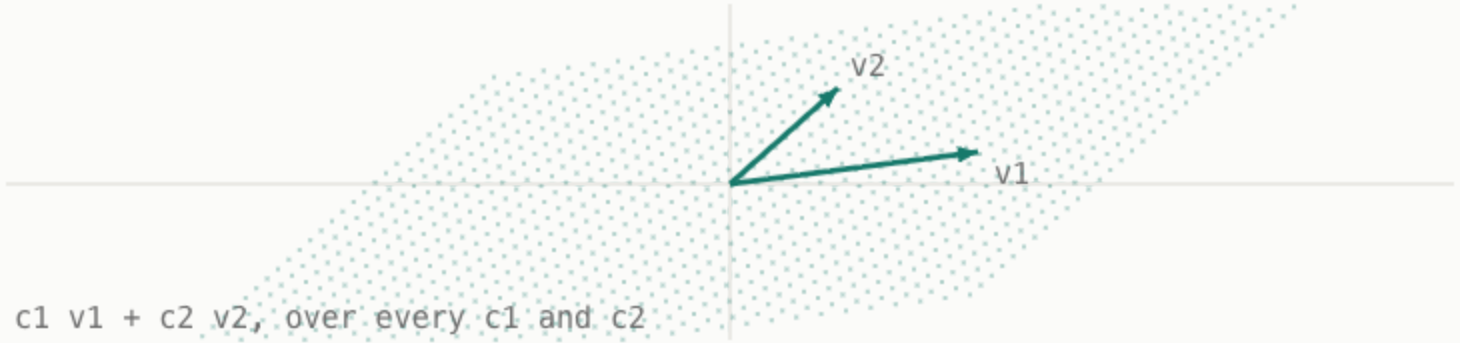
Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.2 linear combination, p. 28. Springer, 2024.

Span and column space

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

SPAN $\operatorname{span}(v_1, \ldots, v_k) = \{c_1 v_1 + \cdots + c_k v_k \ : \ c_i \in \mathbb{R}\}$

{ } every w the weights can reach, all at once



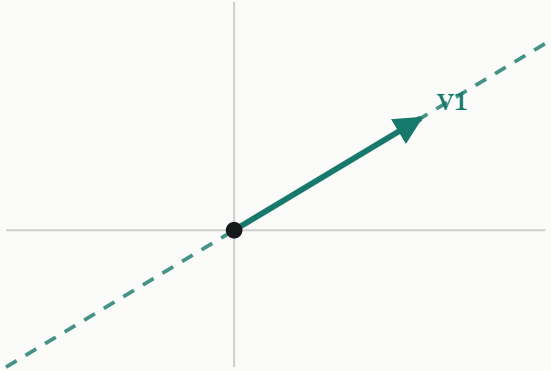
Let both weights roam: the plane fills in.

WORKED EXAMPLE

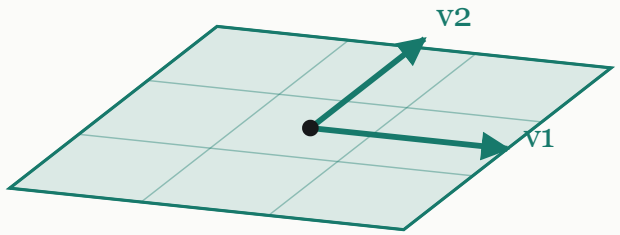
$v_1 = (1, 0, 2)$, $v_2 = (0, 1, 1)$: $3v_1 + 2v_2 = (3, 2, 8)$ is in the span.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.4 span, p. 29. Springer, 2024.

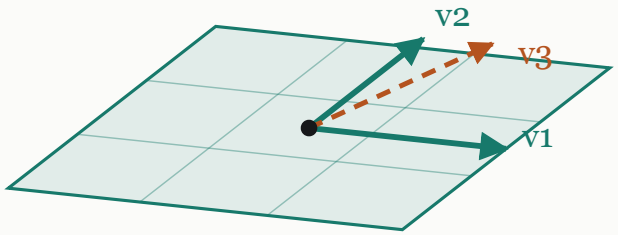
Span and column space



ONE NONZERO VECTOR · A LINE



TWO INDEPENDENT ONES · A PLANE



A THIRD IN IT · SAME PLANE

$$\operatorname{span}(v_1) = \{c\, v_1\}$$
$$\operatorname{span}(v_1, v_2) = \{c_1 v_1 + c_2 v_2\}$$
$$\operatorname{span}(v_1, v_2, v_3) = \operatorname{span}(v_1, v_2)$$

`the conditions` dependent pair: a line, or $\{0\}$ if both zero

`v₃` already a mixture of the first two

Sanderson, G. ***Essence of Linear Algebra***, Chapter 2. 3Blue1Brown, 2016.

Span and column space

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

$$b \in \text{span}(v_1, \ldots, v_k) \iff Vc = b \text{ has a solution}$$

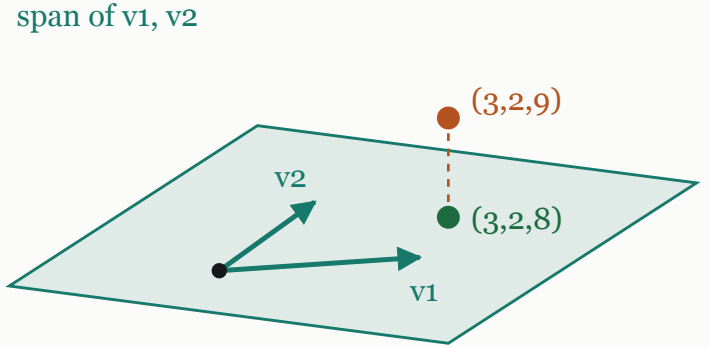
- `V` the arrows as columns
- `c` the unknown weights
- `b` the target vector

IS (3,2,9) IN THE SPAN?

$c_1(1,0,2) + c_2(0,1,1) = (3,2,9)$

row 1 gives $c_1 = 3$, row 2 gives $c_2 = 2$

row 3 is then forced: $2(3) + 2 = 8$, not 9



One lands, one misses.

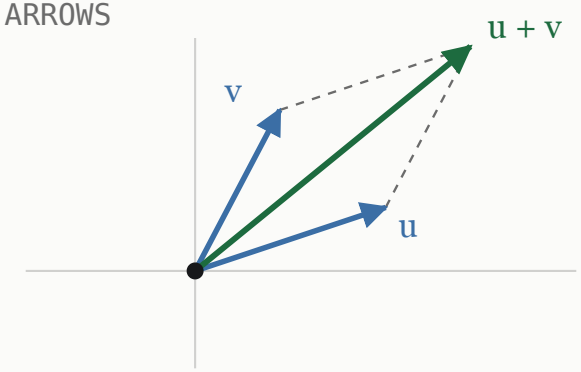
Span and column space

A vector space is a set equipped with vector addition and scalar multiplication satisfying the vector-space axioms.

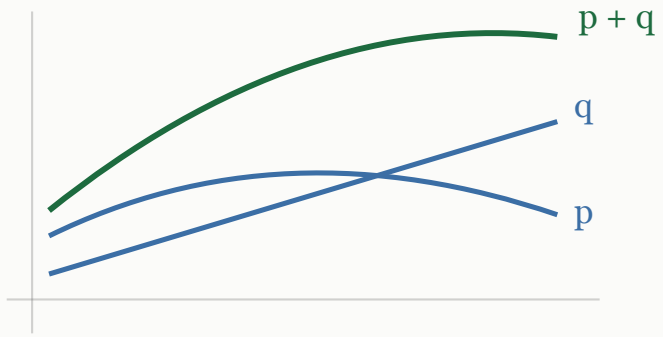
WHAT WE USE $$u, v \in V,\ \alpha, \beta \in \mathbb{R} \implies \alpha u + \beta v \in V$$

`closure` the one property this course leans on all term

ARROWS



POLYNOMIALS



MATRICES

$$\begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} + \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix} = \begin{bmatrix} \bullet & \bullet \\ \bullet & \bullet \end{bmatrix}$$

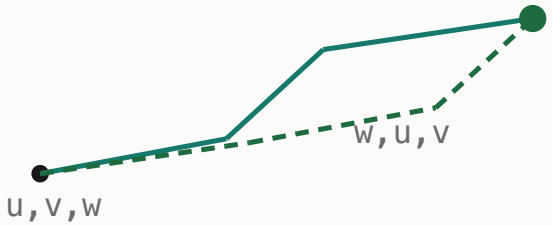
Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.20 vector space, p. 12. Springer, 2024.

Span and column space

ADDITION

$$u+v=v+u \qquad (u+v)+w=u+(v+w)$$
$$v+0=v \qquad \exists w:\ v+w=0$$

`0` and w an origin, and an undo for every vector



ANY ORDER, SAME POINT

SCALING

$$1v=v \qquad \alpha(\beta v)=(\alpha\beta)v$$
$$\alpha(u+v)=\alpha u+\alpha v \qquad (\alpha+\beta)v=\alpha v+\beta v$$

`the payoff` obey all eight and every theorem here applies

Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.20 vector space, p. 12. Springer, 2024.

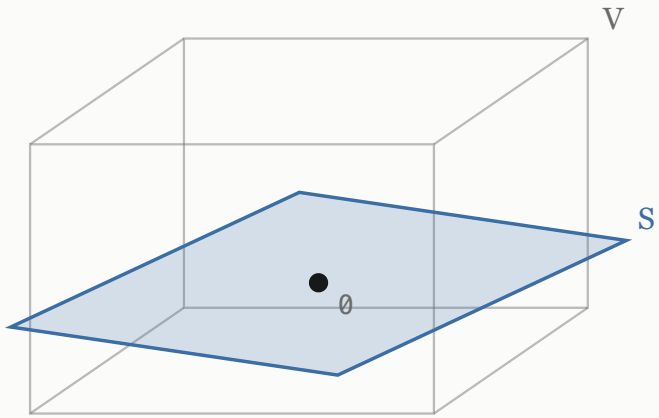
Span and column space

A subset $S \subseteq V$ is a subspace if it is itself a vector space under the same operations.

EQUIVALENT TEST $\boxed{S \neq \varnothing \text{ and } u, v \in S,\ \alpha, \beta \in \mathbb{R} \Rightarrow \alpha u + \beta v \in S}$

`nonempty` plus closure already gives $0 \in S$: choose $\alpha = \beta = 0$

`next page` the same test, split into three checks

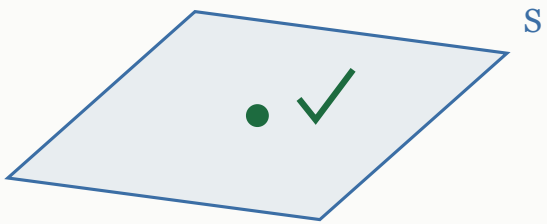


Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.33 subspace and 1.34 conditions for a subspace, p. 18. Springer, 2024.

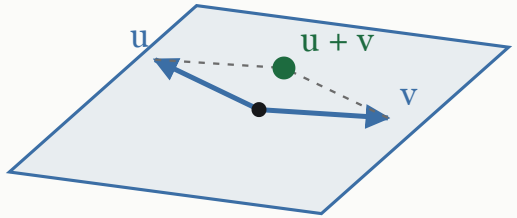
Span and column space

CONDITIONS FOR A SUBSPACE $0 \in S; \qquad u + v \in S; \qquad \alpha u \in S$

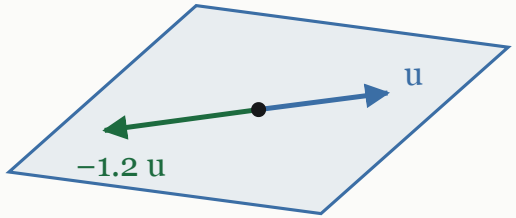
`0`	the zero vector is a member
`u+v`	any two members add to a member
`alpha u`	any multiple stays in, negatives included



0 IS IN S



u + v IS IN S



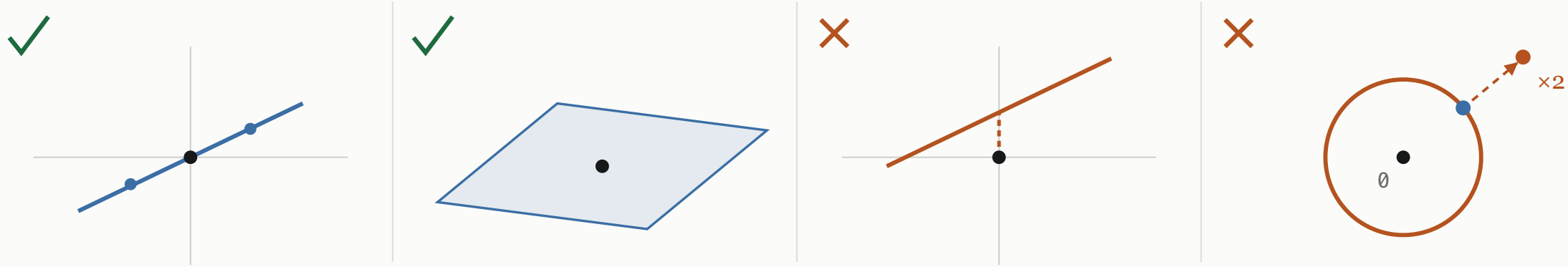
ANY MULTIPLE IS IN S

Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.34 conditions for a subspace, p. 18. Springer, 2024.

Span and column space

FASTEST DISQUALIFIER $0 \notin S \implies S$ is not a subspace

`0 ∈ S` necessary, never sufficient: closure still has to hold



LINE THROUGH 0 · PLANE THROUGH 0 · SHIFTED LINE · AFFINE · CIRCLE

Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.35 example: subspaces, p. 19. Springer, 2024.

Span and column space

CLAIM $v_1, \ldots, v_k \in V \implies \operatorname{span}(v_1, \ldots, v_k)$ is a subspace of V

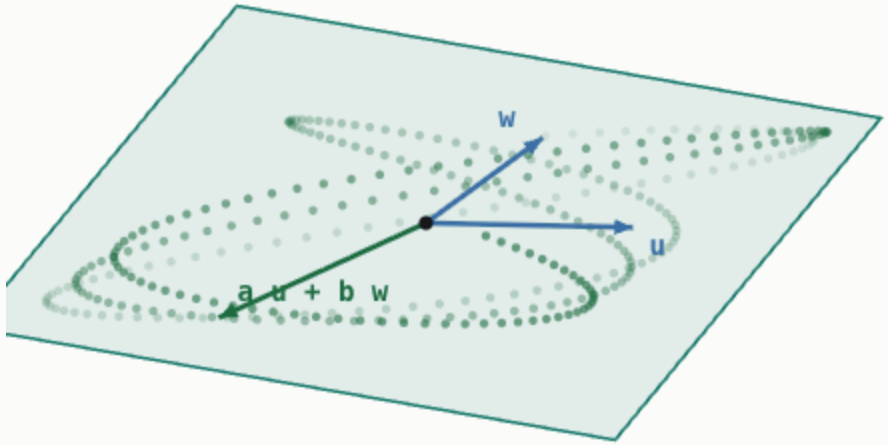
`S` every mixture of the list; the smallest subspace containing it

$$0 = \sum_{i=1}^{k} 0\, v_i \in S$$

`every c_i = 0` the origin is a member

$$\alpha u + \beta w = \sum_{i=1}^{k} (\alpha c_i + \beta d_i)\, v_i \in S$$

`u, w` any two members: $u = \sum_i c_i v_i$, $w = \sum_i d_i v_i$

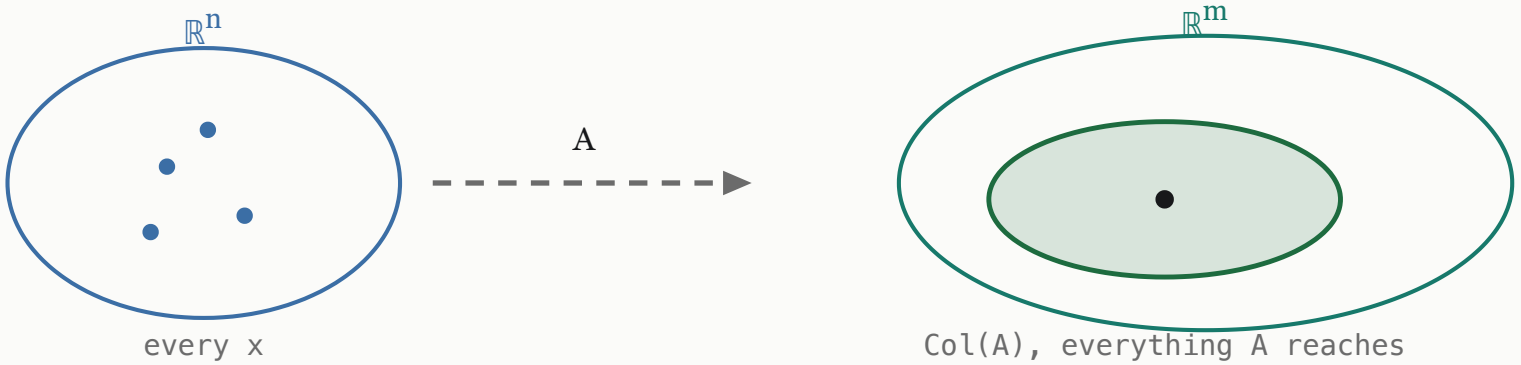


Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.6 span is the smallest containing subspace, p. 29. Springer, 2024.

Span and column space

COLUMN SPACE $$\boxed{\mathrm{Col}(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m}$$

A	any $m \times n$ matrix; x runs over the whole of $\mathbb{R}^n$
$\subseteq \mathbb{R}^m$	a subspace of the codomain; if $m = n$ the roles still differ
also called	the image of A, or the range of A



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range and 3.18 the range is a subspace, p. 61. Springer, 2024.

Span and column space

DEFINITION

$$\mathrm{Col}(A) = \{Ax : x \in \mathbb{R}^n\}$$

`x` runs over every input

COLUMN VIEW

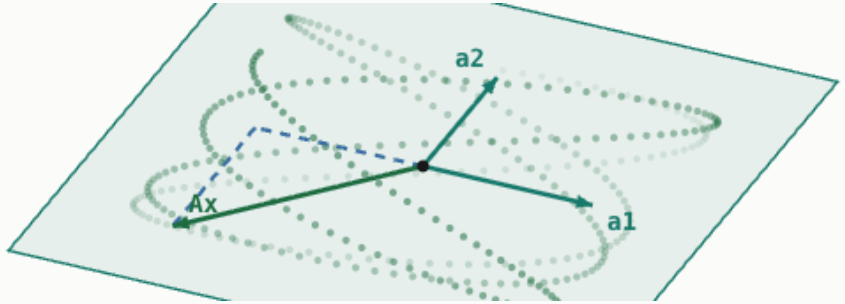
$$Ax = \sum_{j=1}^{n} x_j \, a_j$$

a_j column j of A

⇓

$$\boxed{\mathrm{Col}(A) = \mathrm{span}(a_1, \ldots, a_n)}$$

`one set` reachable outputs and mixtures of the columns are the same thing



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range, p. 61; 3.50 linear combination of columns, p. 76. Springer, 2024.

Span and column space

QUESTION $Ax = b$?

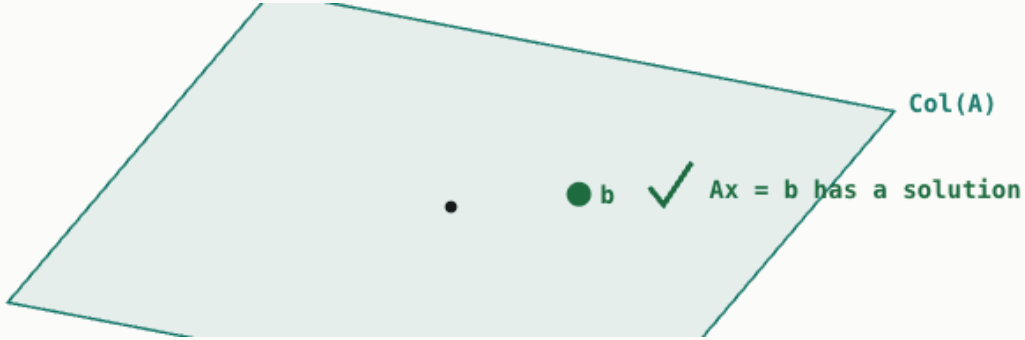
`b` a target on the output side

$$Ax = b \text{ has a solution} \iff \exists\, x :\ Ax = b \iff b \in \{Ax : x \in \mathbb{R}^n\}$$

`∃ x` the whole question is whether one input reaches b

$$\boxed{Ax = b \text{ is solvable} \iff b \in \mathrm{Col}(A)}$$

`nothing new` the right side is the definition of Col(A), read backwards



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range, p. 61. Springer, 2024.

Span and column space

SOLVE $c_1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ y_3 \end{bmatrix}$

row 1 $c_1 + 0 = 3$, so $c_1 = 3$

row 2 $0 + c_2 = 2$, so $c_2 = 2$

row 3 $2c_1 + c_2 = 2(3) + 2 = 8$ must equal y_3

TARGET 1

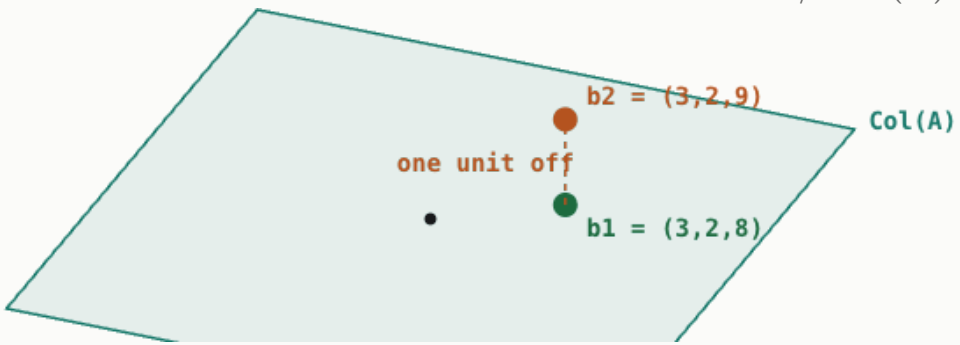
$y_3 = 8 : \; 8 = 8$ ✓

so $b_1 = 3a_1 + 2a_2 \in \mathrm{Col}(A)$

TARGET 2

$y_3 = 9 : \; 9 \neq 8$ ×

so no c works: $b_2 \notin \mathrm{Col}(A)$



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range, p. 61. Springer, 2024.

Span and column space

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\boxed{Ax = \sum_j x_j \, a_j}$$

x_j the amount taken of column j

$\Downarrow$

$$\boxed{\mathrm{Col}(A) = \mathrm{span}(a_1, \ldots, a_n)}$$

span every mixture, so every reachable output

$\Downarrow$

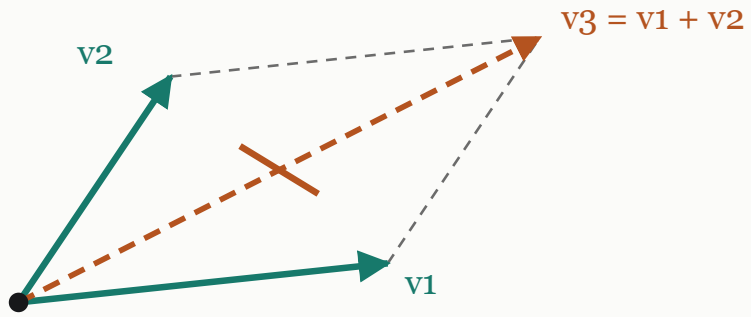
$$\boxed{Ax = b \text{ solvable} \iff b \in \mathrm{Col}(A)}$$

our A $a_3 = a_1 + a_2$, so the third column adds nothing to the span

> When do different columns carry independent information?

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.15 linearly independent, p. 32. Springer, 2024.

Null Space and Linear Independence



ENOUGH TO REACH?

OR ONE TOO MANY?

Null Space and Linear Independence

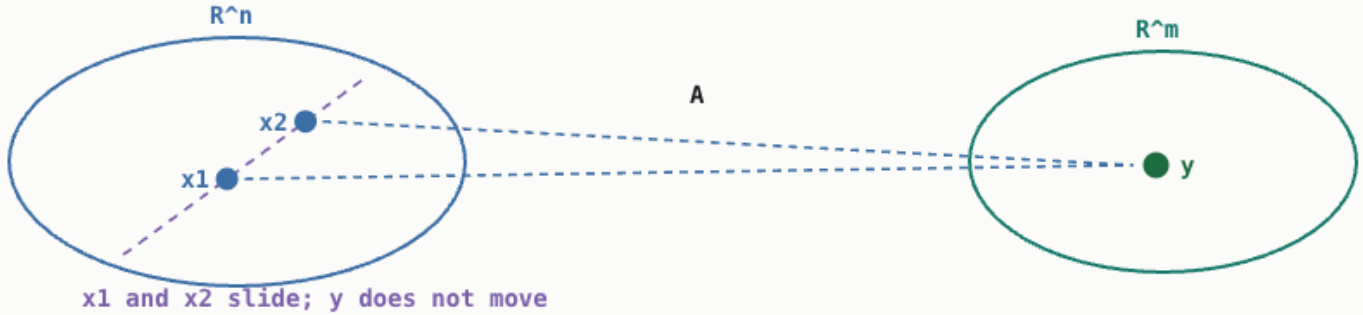
SO FAR $\mathrm{Col}(A) =$ all reachable outputs

settled which targets can be hit, and by which mixtures

NOW ASK $\boxed{Ax_1 = Ax_2, \qquad x_1 \neq x_2}$

one output but more than one input that reaches it

> How can two different inputs produce the same output?



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.14 injective, p. 60; 3.11 null space, p. 59. Springer, 2024.

Null Space and Linear Independence

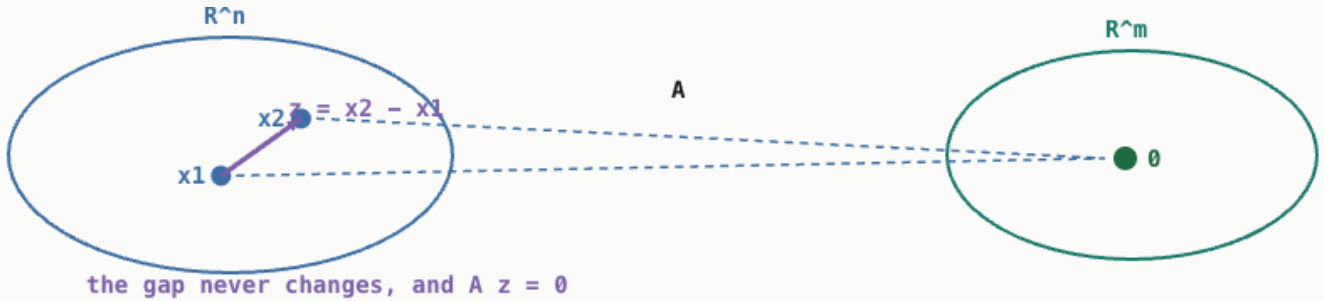
$$Ax_1 = Ax_2$$
$$Ax_1 - Ax_2 = 0$$
$$A(x_1 - x_2) = 0$$

`last step` linearity, and nothing else, is what pulls A outside

THE DIFFERENCE $\boxed{z = x_1 - x_2 \implies Az = 0}$

`z` nonzero, since $x_1 \neq x_2$, and A sends it to zero

> Every collision $Ax_1 = Ax_2$ is a solution $z = x_1 - x_2$ of $Ax = 0$.



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.1 linear map, p. 52; 3.11 null space, p. 59. Springer, 2024.

Null Space and Linear Independence

HOMOGENEOUS SYSTEM $\boxed{Ax = 0}$

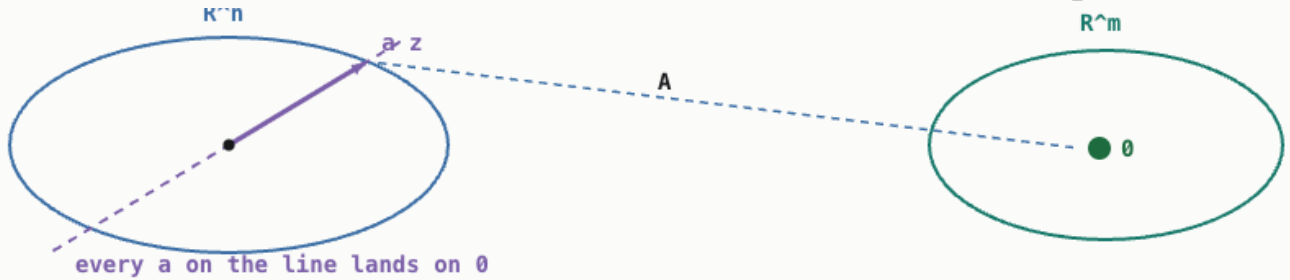
`homogeneous` the right-hand side is the zero vector

`x = 0` always a solution, and the uninteresting one

> Are there nonzero solutions?

ONE GIVES A LINE $\boxed{Az = 0,\ z \neq 0 \implies A(\alpha z) = 0\ \forall \alpha \in \mathbb{R}}$

`αz` one invisible direction, so a whole line of invisible inputs



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.26 homogeneous system of linear equations, p. 65. Springer, 2024.

Null Space and Linear Independence

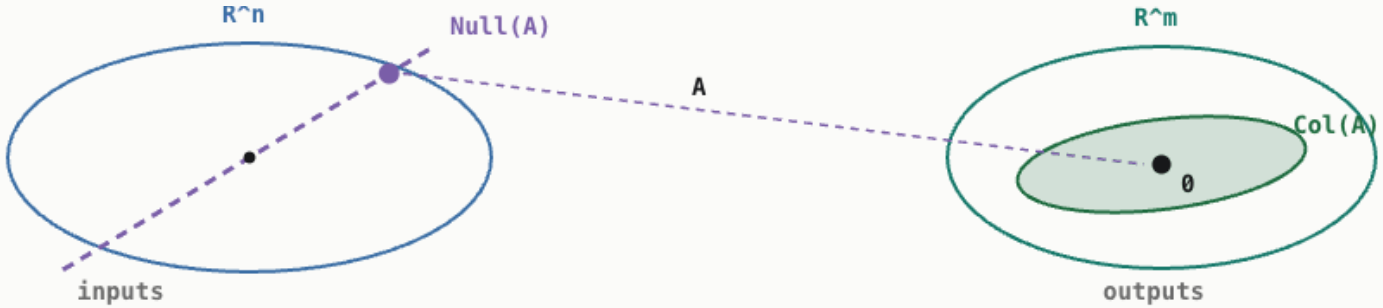
NULL SPACE $\boxed{\text{Null}(A) = \{x \in \mathbb{R}^n : Ax = 0\}}$

also called the kernel of A, written $\ker(A)$

$Ax = 0$ the inputs A cannot see

SHAPE $\boxed{\text{Null}(A) \subseteq \mathbb{R}^n}$ against $\text{Col}(A) \subseteq \mathbb{R}^m$

input side Null(A) lives where x lives, Col(A) where Ax lives



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space and 3.13 the null space is a subspace, p. 59. Springer, 2024.

Null Space and Linear Independence

$$u, v \in \text{Null}(A) \implies Au = 0,\ Av = 0$$

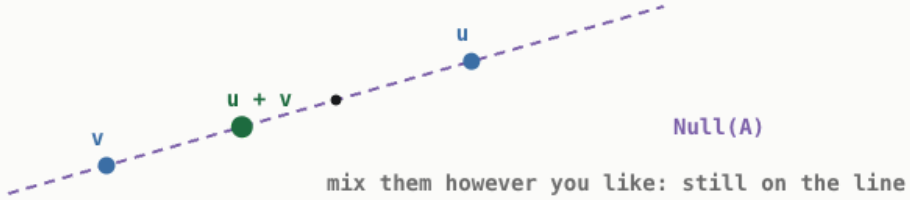
`and` $A0 = 0$, so the origin is in $\text{Null}(A)$ to begin with

$$A(\alpha u + \beta v) = \alpha Au + \beta Av = 0$$

`so` $\alpha u + \beta v \in \text{Null}(A)$ for every α, β

`SUBSPACE` $\boxed{\text{Null}(A) \text{ is a subspace of } \mathbb{R}^n}$

`flat` in $\mathbb{R}^3$: $\{0\}$, a line, a plane, or all of $\mathbb{R}^3$



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.13 the null space is a subspace, p. 59. Springer, 2024.

Null Space and Linear Independence

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

OUR MATRIX $a_3 = a_1 + a_2 \implies a_1 + a_2 - a_3 = 0$

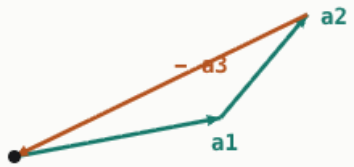
read it a mixture of the columns, weights $1, 1, -1$, giving zero

$$A \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = 0 \implies \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \in \mathrm{Null}(A)$$

and next: solve for the whole set

THE WHOLE SET $\boxed{\mathrm{Null}(A) = \mathrm{span}\left\{(1, 1, -1)^{\mathsf{T}}\right\}}$

a line through the origin



a1 + a2 − a3 walks back to 0

so (1, 1, −1) is in Null(A)

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Null Space and Linear Independence

$$\text{Ax = 0} \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

row 1 $1x_1 + 0x_2 + 1x_3 = 0$, so $x_1 + x_3 = 0$

row 2 $0x_1 + 1x_2 + 1x_3 = 0$

row 3 $2x_1 + x_2 + 3x_3 = 0$ is twice row 1 plus row 2

SOLVE $x_3 = t:\ x_1 = -t,\ x_2 = -t,\ x = t(-1, -1, 1)^{\mathsf{T}}$

why one free unknown

NULL SPACE $\boxed{\text{Null}(A) = \text{span}\{(-1, -1, 1)^{\mathsf{T}}\}}$

check $-1 + 1,\ -1 + 1,\ -2 - 1 + 3$: all zero

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Null Space and Linear Independence

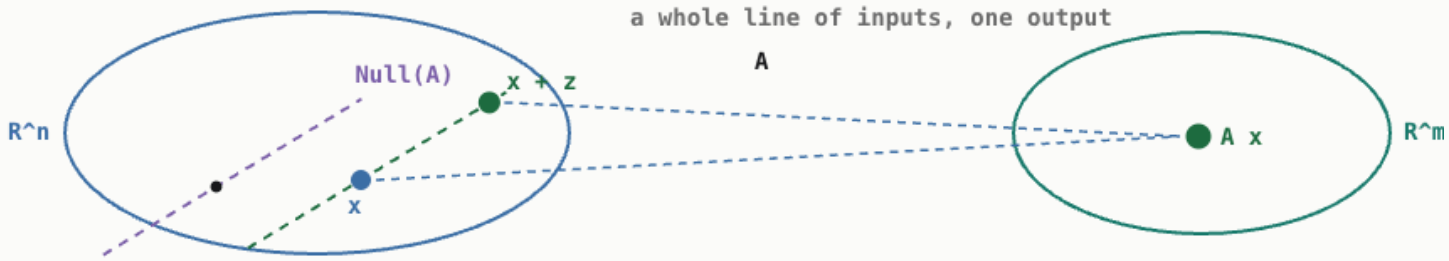
$$z \in \text{Null}(A) \implies Az = 0 \implies A(x+z) = Ax + Az = Ax$$

`x` any input at all, not a special one

WHAT IT MEANS $\boxed{A(x+z) = Ax, \qquad z \in \text{Null}(A)}$

`x + z` for our matrix, a line of inputs sharing one output

> Null-space directions are invisible to the map.



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.1 linear map, p. 52; 3.11 null space, p. 59. Springer, 2024.

Null Space and Linear Independence

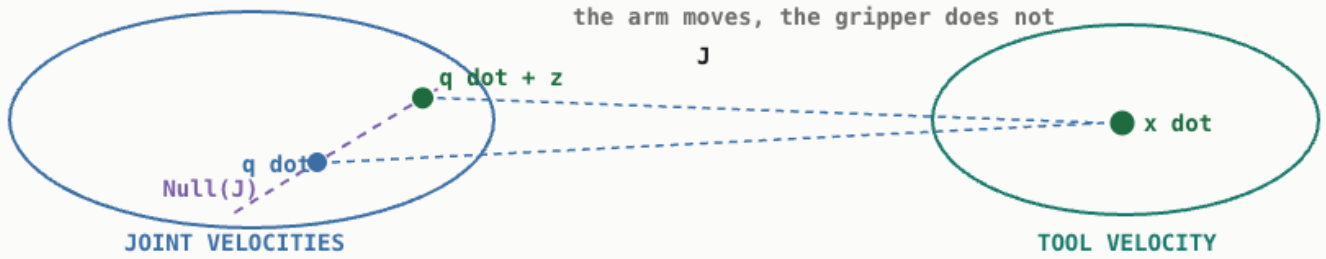
ON A ROBOT $\dot{x} = J(q)\,\dot{q}$

$\dot{q}$ joint velocities, what you command

$\dot{x}$ end-effector velocity, what the world sees

REDUNDANCY $\boxed{z \in \mathrm{Null}(J(q)) \implies J(q)(\dot{q} + z) = J(q)\dot{q}}$

z a joint-velocity direction that produces no end-effector velocity



Lynch, K. M. and Park, F. C. ***Modern Robotics***, §6.3 inverse velocity kinematics, p. 233. Cambridge University Press, 2017 preprint.

Null Space and Linear Independence

LINEAR INDEPENDENCE $$c_1 v_1 + \cdots + c_k v_k = 0 \implies c_1 = \cdots = c_k = 0$$

`read it` the all-zero weights are the only ones that cancel

SAME THING $$\sum_{i=1}^{k} c_i v_i = 0 \implies c = 0$$

`c` the whole list of weights, read as one vector



DEPENDENT · THE WALK CLOSES

INDEPENDENT · IT NEVER CLOSES

Left, a nonzero walk comes home. Right, none does.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.15 linearly independent, p. 32. Springer, 2024.

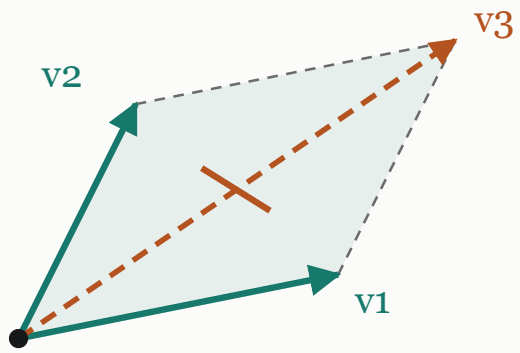
Null Space and Linear Independence

LINEAR DEPENDENCE $$\boxed{\exists\, c \neq 0 :\ c_1 v_1 + \cdots + c_k v_k = 0}$$

$c \neq 0$ at least one weight is nonzero; say it is c_j

SOLVE FOR THAT ONE $$\boxed{v_j = -\frac{1}{c_j} \sum_{i \neq j} c_i v_i}$$

so v_j is built from the others, and deleting it leaves the span unchanged



DELETE v3

SAME PLANE

Strike the diagonal and the plane is untouched.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.17 linearly dependent and 2.19 linear dependence lemma, p. 33. Springer, 2024.

Null Space and Linear Independence

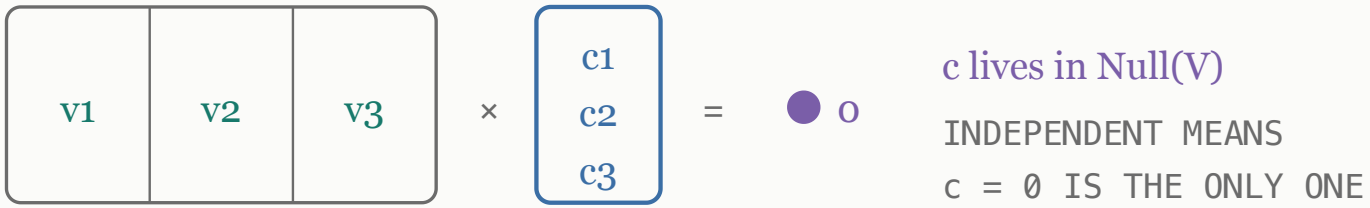
STACK THEM $V = [\, v_1 \ \cdots \ v_k \,], \qquad Vc = \sum_i c_i v_i$

`Vc` the matrix times its weights

THE ANCHOR

$$\begin{aligned} v_1, \ldots, v_k \text{ independent} &\iff \text{Null}(V) = \{0\} \\ v_1, \ldots, v_k \text{ dependent} &\iff \text{Null}(V) \neq \{0\} \end{aligned}$$

`one question` redundancy in a list is a null space



c lives in Null(V)

INDEPENDENT MEANS
c = 0 IS THE ONLY ONE

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.15 linearly independent, p. 32; 3.15 injectivity, p. 60. Springer, 2024.

Null Space and Linear Independence

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

OUR MATRIX $a_3 = a_1 + a_2 \implies A(1, 1, -1)^\mathsf{T} = 0$

one equation the weights $1, 1, -1$ cancel the three columns

NULL SPACE

$$(1, 1, -1)^\mathsf{T} \in \mathrm{Null}(A)$$

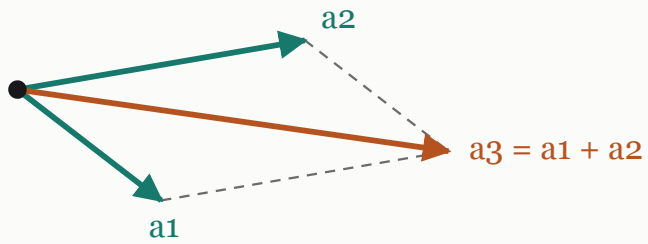
reading an input A cannot see

LINEAR DEPENDENCE

$$a_1, a_2, a_3 \text{ dependent}$$

reading a column you may delete

> Same equation, two interpretations.



THE THIRD COLUMN
BUYS NOTHING

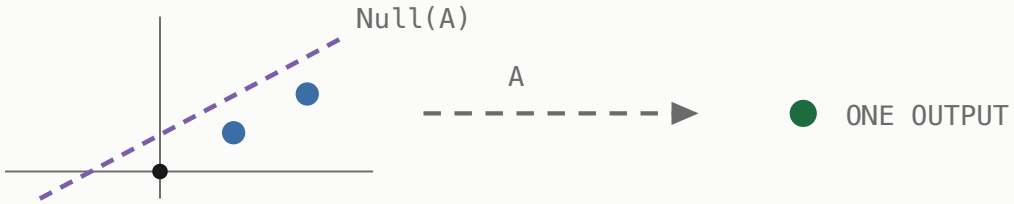
Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.17 linearly dependent, p. 33; 3.11 null space, p. 59. Springer, 2024.

Null Space and Linear Independence

$$\begin{aligned} \text{Null}(A) &= \{x : Ax = 0\} \\ z \in \text{Null}(A) &\implies A(x+z) = Ax \\ \text{Null}(A) \neq \{0\} &\implies \text{information is lost} \end{aligned}$$

`the chain` definition, meaning, cost

FOR THE COLUMNS $\text{Null}(A) = \{0\} \iff$ columns independent

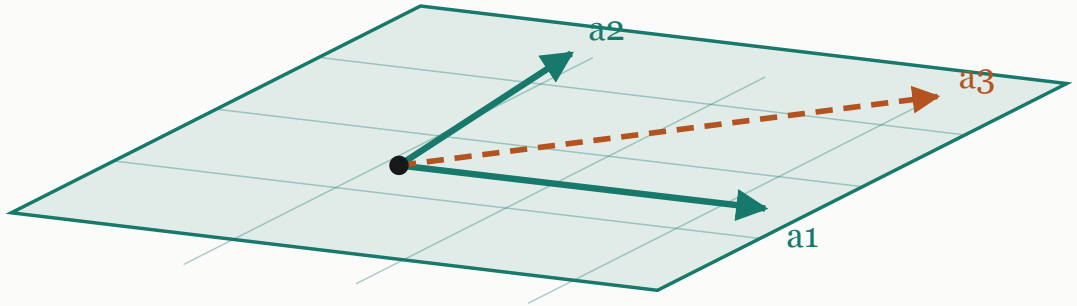


`settled` whether a direction is redundant

> How many independent directions?

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.26 basis, p. 39. Springer, 2024.

Basis, Dimension, and Rank



HOW MANY

SURVIVE?

Basis, Dimension, and Rank

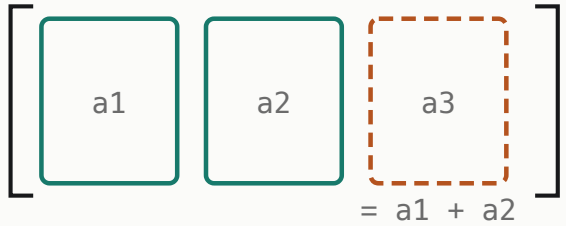
$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

WE KNOW $\mathrm{Col}(A)$ what A can produce, $\mathrm{Null}(A)$ what A cannot see

and how to tell whether a direction is redundant

NOW ASK **How many independent directions are needed?**

our A $a_3 = a_1 + a_2$: three columns, but how many directions?



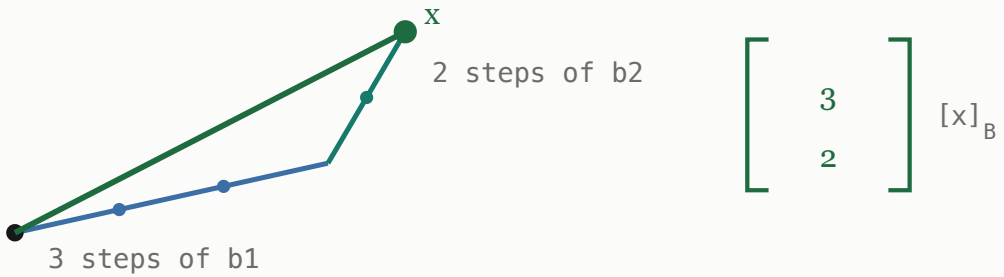
THREE COLUMNS

HOW MANY DIRECTIONS?

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.26 basis, p. 39. Springer, 2024.

Basis, Dimension, and Rank

BASIS $\boxed{\text{Basis} = \text{Spanning} + \text{Linear Independence}}$



`spans` $\operatorname{span}(b_1, \ldots, b_k) = V$

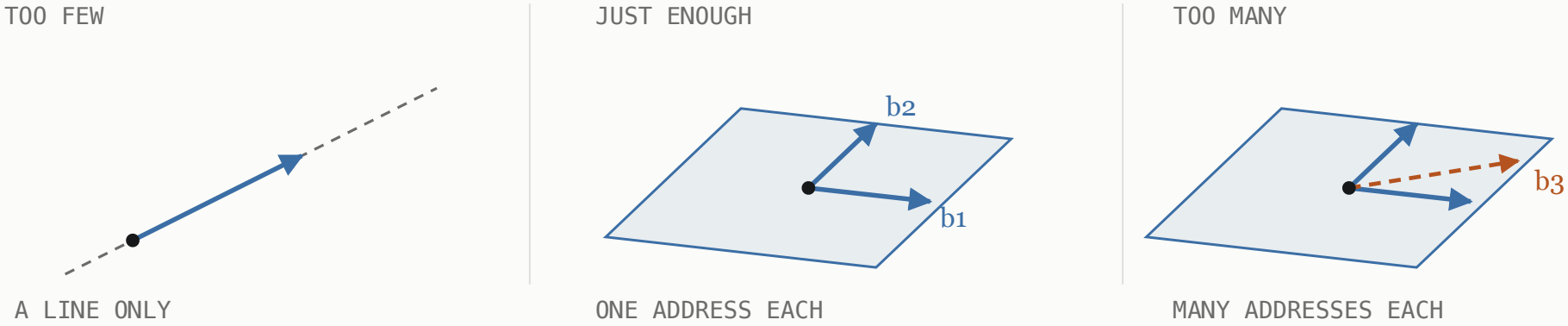
`independent` $c_1 b_1 + \cdots + c_k b_k = 0 \Rightarrow c_1 = \cdots = c_k = 0$

SO enough to build everything, with nothing spare

`both halves` either one alone is not a basis

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.26 basis, p. 39. Springer, 2024.

Basis, Dimension, and Rank



Too few cannot reach; too many give several addresses.

BASIS enough, but not too many

`too few` here independent, but does not reach everything

`too many` reaches everything, but carries a repeat

WORKED EXAMPLE

In $\mathbb{R}^2$: $\{(1,0)\}$ is too few, $\{(1,0),(1,1)\}$ is a basis, adding $(2,1)$ is too many.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.26 basis, p. 39. Springer, 2024.

Basis, Dimension, and Rank

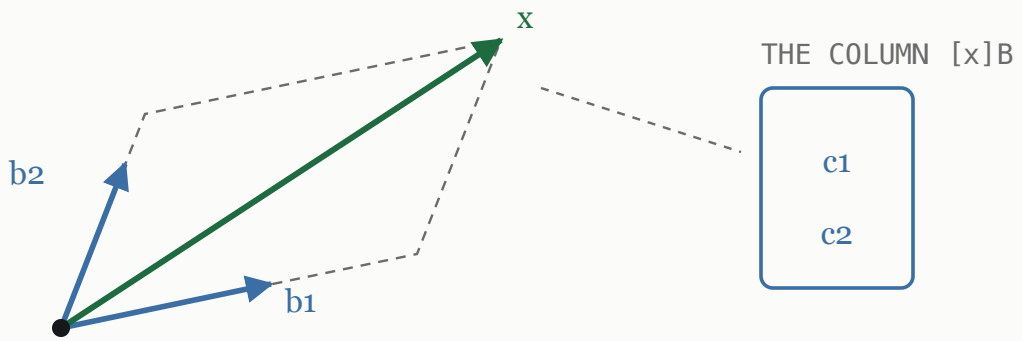
UNIQUE WEIGHTS $x = c_1 b_1 + \cdots + c_k b_k$ in exactly one way

why independence: two different weightings would differ by a nonzero mix that vanishes

COORDINATES

$$[x]_B = (c_1, \ldots, c_k)^{\mathsf{T}}, \qquad x = B\,[x]_B$$

one address the vector does not change; only its coordinates do



Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.28 criterion for basis, p. 39; 3.73 matrix of a vector, p. 88. Springer, 2024.

Basis, Dimension, and Rank

DIMENSION $\dim V =$ the number of vectors in any basis of V



one more independent direction each time

`any` every basis of the same space has the same length

READING dimension counts independent degrees of freedom

`finite` every space in this course has a finite basis

$$\dim\{0\} = 0$$

`zero space` the empty list is a basis of it

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.35 dimension, p. 44. Springer, 2024.

Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$



BASIS b BASIS d

One plane, two bases, both of size two.

SAME SIZE ALWAYS

different bases, different arrows, the same number of them

`this plane` every basis of it has exactly two vectors

`so` dimension belongs to the space, not to a choice

Axler, S. ***Linear Algebra Done Right***, 4th ed., 2.35 dimension, p. 44. Springer, 2024.

Basis, Dimension, and Rank

REDUCE $$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \xrightarrow{r_3 \leftarrow r_3 - 2r_1} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \xrightarrow{r_3 \leftarrow r_3 - r_2} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = R$$

`step 1` $(2,1,3) - 2(1,0,1) = (0,1,1)$

`step 2` $(0,1,1) - (0,1,1) = (0,0,0)$

`pivots` columns 1 and 2

WHY R $R = EA$, E invertible : $Ax = 0 \iff Rx = 0$

`so` $\text{Null}(A) = \text{Null}(R)$

Strang, G. ***18.06***, Lecture 7: pivot variables and special solutions. MIT OCW, 2010.

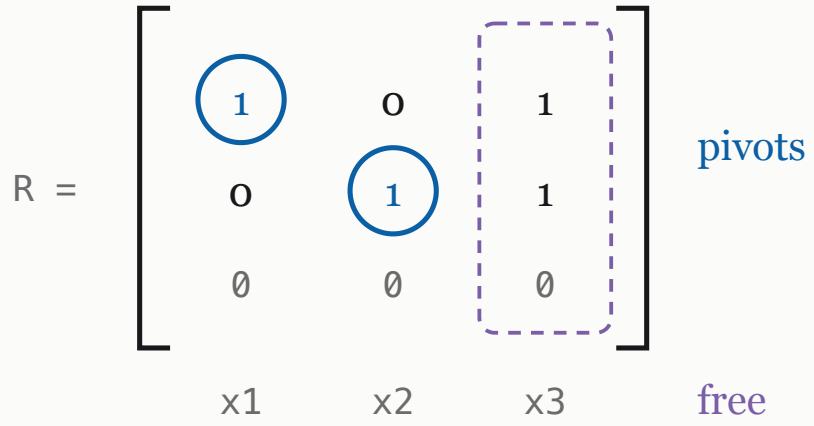
Basis, Dimension, and Rank

PIVOT the leading nonzero entry of a nonzero row of R

`in R` pivot positions are read off the echelon form, not off A

`pivot variable` a variable whose column carries a pivot

`free variable` a variable whose column does not



HERE x_1, x_2 pivot; x_3 free

`count` 3 variables, 2 pivots, 1 free

Strang, G. ***18.06***, Lecture 7: pivot variables and special solutions. MIT OCW, 2010.

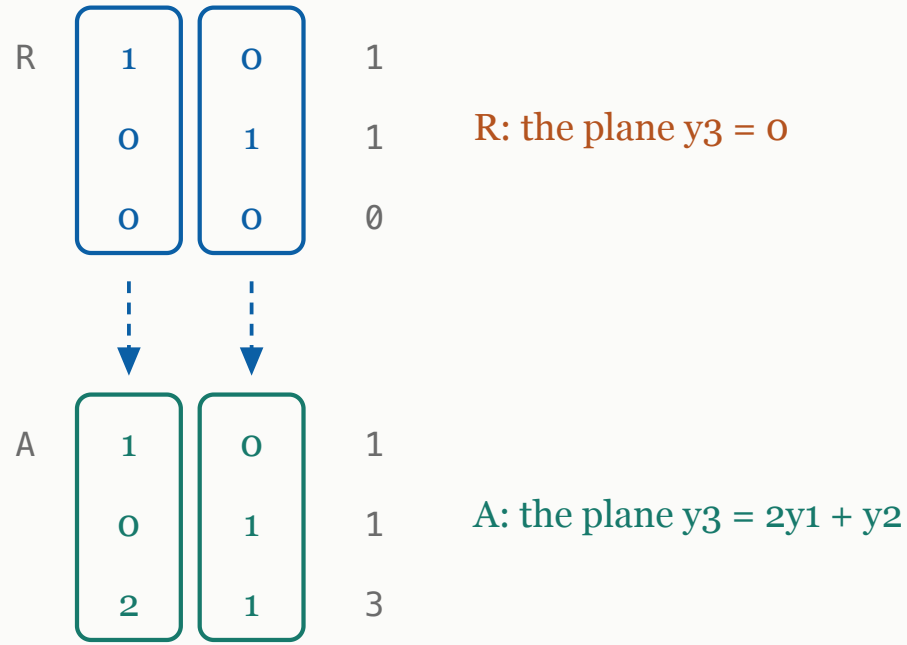
Basis, Dimension, and Rank

PIVOT COLUMNS | the pivot columns of A are a basis for $\mathrm{Col}(A)$

here $\mathrm{Null}(A) = \mathrm{Null}(R)$, while $\mathrm{Col}(A) \neq \mathrm{Col}(R)$

R's columns $(1,0,0)^{\mathsf{T}}, (0,1,0)^{\mathsf{T}}$ span $y_3 = 0$; $a_1 = (1,0,2)^{\mathsf{T}}$ has $y_3 = 2$

A's columns a_1, a_2 independent, and $a_3 = a_1 + a_2$, so they span $\mathrm{Col}(A)$



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.52 column rank, p. 77. Springer, 2024.

Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix} \quad R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

ROWS OF R $x_1 + x_3 = 0, \qquad x_2 + x_3 = 0, \qquad 0 = 0$

`legal` $\mathrm{Null}(A) = \mathrm{Null}(R)$

`free` column 3 has no pivot: $x_3 = t$

SOLVE $x_1 = -t, \quad x_2 = -t \implies x = (-t, -t, t)^{\mathsf{T}} = t(-1, -1, 1)^{\mathsf{T}}$

$t = 1$ on A: $-1 + 1,\ -1 + 1,\ -2 - 1 + 3$, all zero

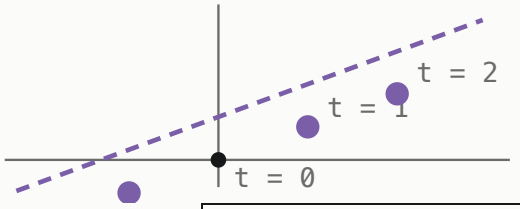
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Basis, Dimension, and Rank

ALL SOLUTIONS $\mathrm{Null}(A) = \{\, t(-1,-1,1)^{\mathsf{T}} : \; t \in \mathbb{R} \,\}$

why every solution is t times one vector, nothing else

so all multiples of one vector: a span



BASIS AND DIMENSION $\mathrm{Null}(A) = \mathrm{span}\{(-1,-1,1)^{\mathsf{T}}\}, \qquad \dim \mathrm{Null}(A) = 1$

basis one nonzero vector is independent, hence a basis

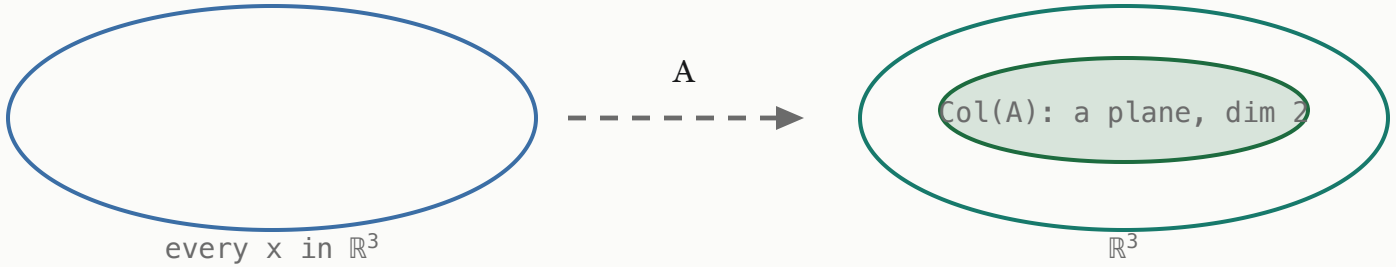
count one free variable, one basis vector: $\dim \mathrm{Null}(A)$ = free variables

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59; 2.26 basis, p. 39. Springer, 2024.

Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

RANK $\boxed{\operatorname{rank}(A) = \dim \operatorname{Col}(A)}$



our A $\dim \operatorname{Col}(A) = 2$, so $\operatorname{rank}(A) = 2$

TO COMPUTE $\boxed{\operatorname{rank}(A) = \text{the number of pivots}}$

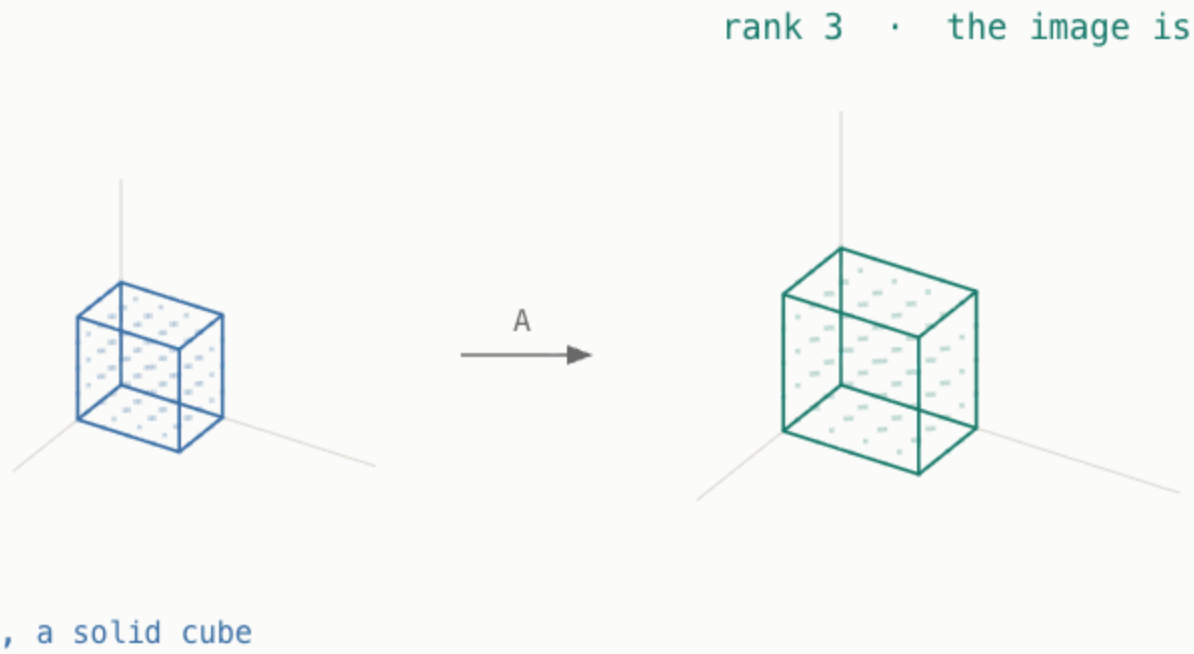
reading how many independent output directions A can produce

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.58 rank, p. 79; 3.52 column rank, p. 77. Springer, 2024.

Basis, Dimension, and Rank

INTO THREE-SPACE $r = 3$ volume; $r = 2$ plane; $r = 1$ line; $r = 0$ the point $\{0\}$

`rank` the dimension of the reachable output space



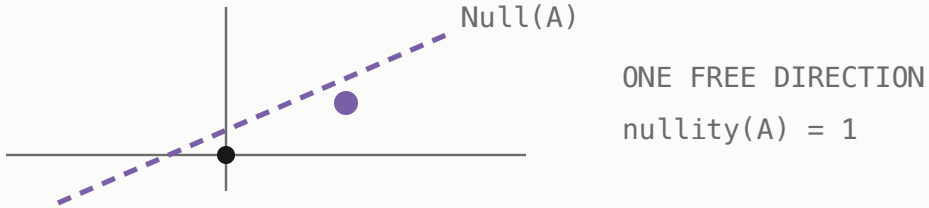
The input cube never changes. Each direction the matrix loses costs the image a whole dimension.

Sanderson, G. ***Essence of Linear Algebra***, ch. 7. 3Blue1Brown, 2016.

Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

NULLITY $\boxed{\text{nullity}(A) = \dim \text{Null}(A)}$



our A $\text{Null}(A) = \text{span}\{(-1, -1, 1)^T\}$, so $\text{nullity}(A) = 1$

READING independent input directions lost by A

the pair rank counts what survives, nullity counts what disappears

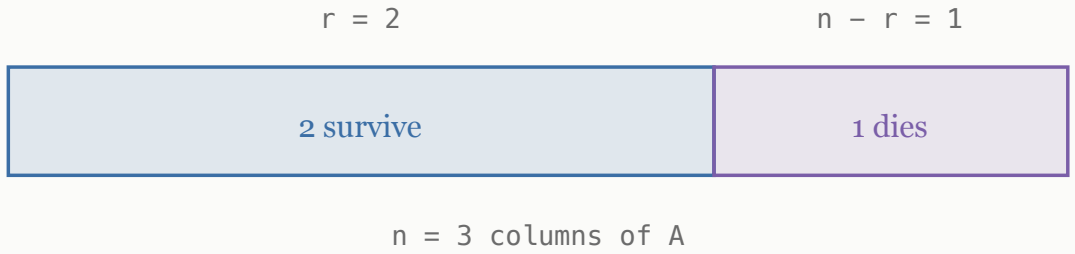
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Basis, Dimension, and Rank

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

RANK–NULLITY $\boxed{\operatorname{rank}(A) + \operatorname{nullity}(A) = n}$

`n`	the number of columns, the input dimension
`reach`	$\dim \operatorname{Col}(A) = r$: directions that survive
`loss`	$\dim \operatorname{Null}(A) = n - r$: directions that disappear
`our A`	$2 + 1 = 3$
`a choice`	$n - r$ basis vectors in $\operatorname{Null}(A)$, r whose images span $\operatorname{Col}(A)$



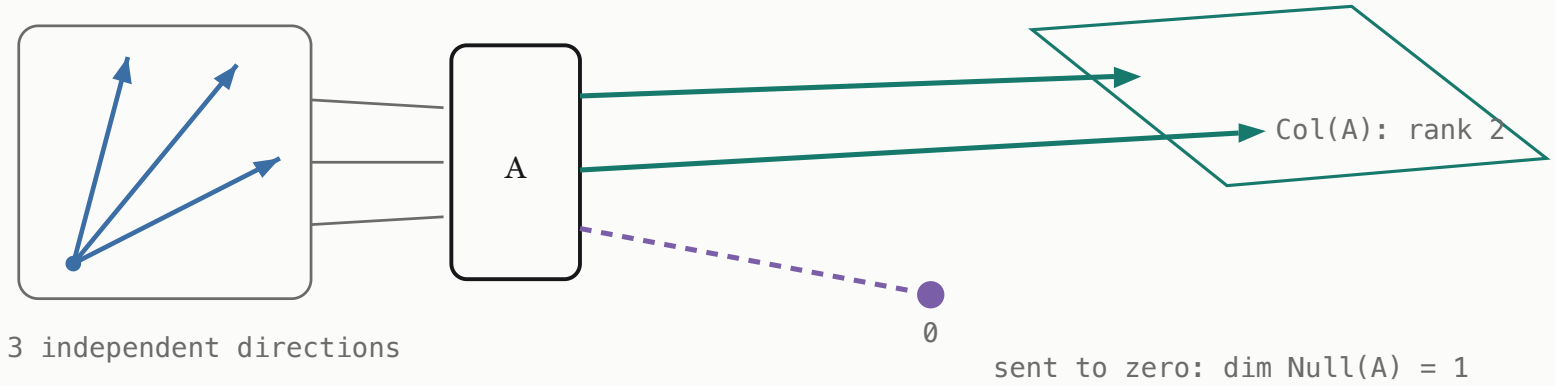
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Basis, Dimension, and Rank

THE MAP $A : \mathbb{R}^n \to \mathbb{R}^m, \qquad n = r + (n - r)$

`r` directions that survive

`n − r` directions that vanish



Two survivors, one casualty, three directions accounted for.

SO **rank–nullity counts the input side**

`not m` the null space is not in the output space

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

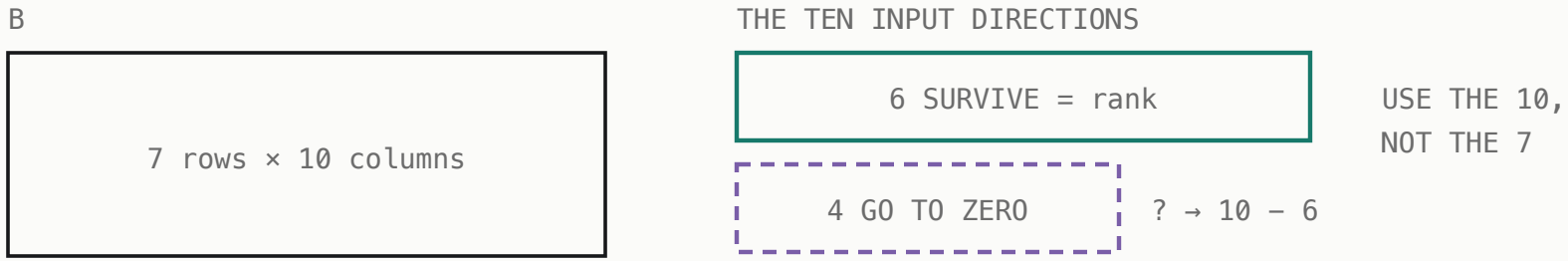
Basis, Dimension, and Rank

EXAM QUESTION $B \in \mathbb{R}^{7\times 10}, \qquad \operatorname{rank}(B) = 6$

ask $\dim \operatorname{Col}(B)$, $\dim \operatorname{Null}(B)$, and the number of free variables

$$\dim \operatorname{Col}(B) = 6, \qquad \dim \operatorname{Null}(B) = 10 - 6 = 4, \qquad 4 \text{ free variables}$$

the trap use the 10, not the 7



$$\dim \operatorname{Null}(B) = n - r = 10 - 6 = 4.$$

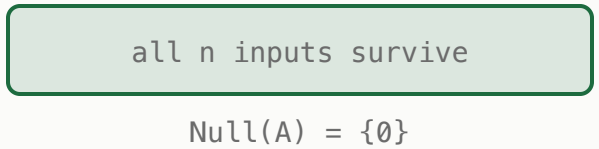
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Basis, Dimension, and Rank

FULL COLUMN RANK

$$r = n$$

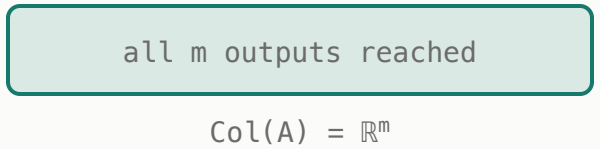
then $\text{Null}(A) = \{0\}$: no input direction is lost; needs $n \leq m$



FULL ROW RANK

$$r = m$$

then $\text{Col}(A) = \mathbb{R}^m$: every output is reachable; needs $m \leq n$



> One is about losing nothing. The other is about reaching everything.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Basis, Dimension, and Rank

WHERE WE ARE $\dim \mathrm{Col}(A) = r, \qquad \dim \mathrm{Null}(A) = n - r$



so the three questions become countable

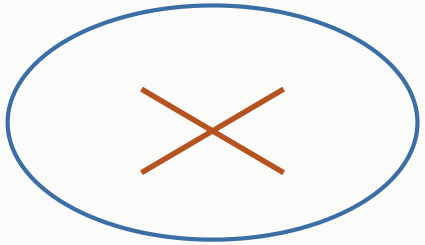
$$\begin{aligned} \text{is } b \text{ reachable?} \quad & b \in \mathrm{Col}(A) \\ \text{is } x \text{ unique?} \quad & \mathrm{Null}(A) = \{0\} \\ \text{how many free directions?} \quad & n - r \end{aligned}$$

row two uniqueness is a question only once a solution exists

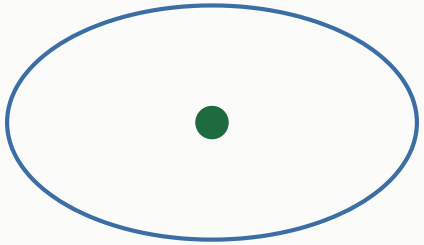
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Linear Systems and Solution Sets

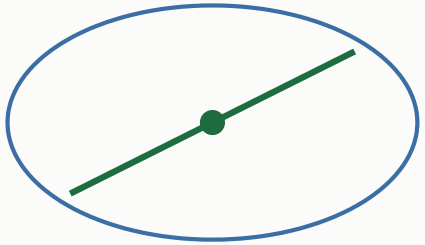
NONE



EXACTLY ONE



INFINITELY MANY



Linear Systems and Solution Sets

RETURN TO $Ax = b$



`1 existence` does a solution exist?

`2 uniqueness` if it exists, is it the only one?

`3 structure` what are all of them?

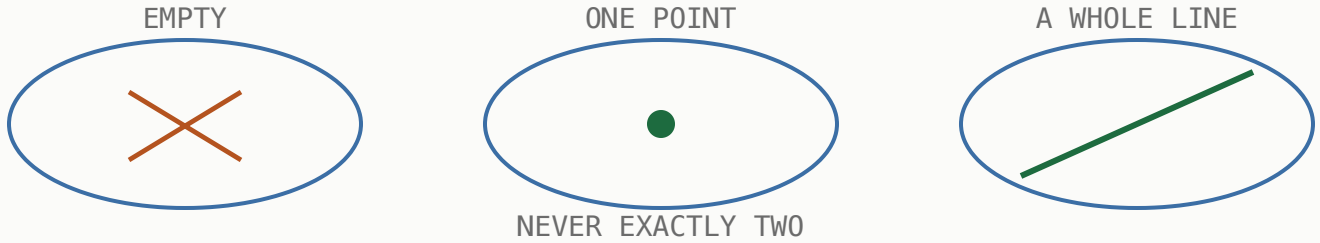
WITH $\mathrm{Col}(A)$ and $\mathrm{Null}(A)$

`nothing new` both spaces are already built and measured

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range, p. 61; 3.11 null space, p. 59. Springer, 2024.

Linear Systems and Solution Sets

SOLUTION SET $\mathcal{S}(A, b) = \{x \in \mathbb{R}^n : Ax = b\}$



`a solution` any x that produces the requested b

`consistent` $\mathcal{S}(A, b) \neq \emptyset$; otherwise inconsistent

THREE POSSIBILITIES empty, one point, infinitely many

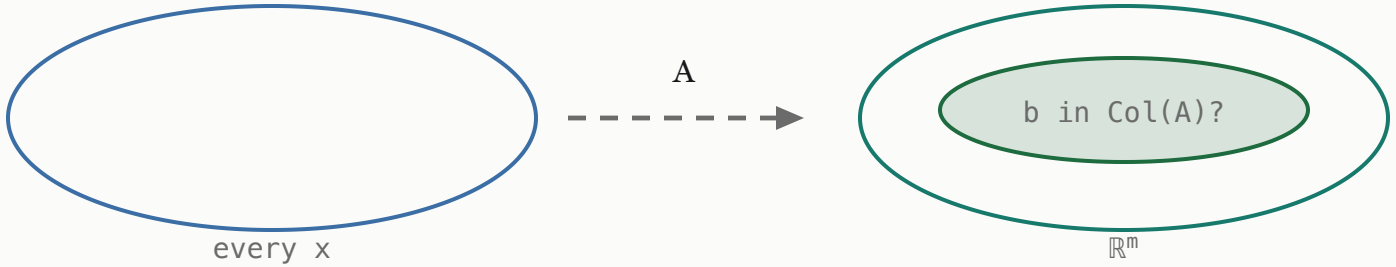
`never two` two solutions force infinitely many: a line, a plane, or higher

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Linear Systems and Solution Sets

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 2 & 1 & 3 \end{bmatrix}$$

RECALL $\mathrm{Col}(A) = \{Ax : x \in \mathbb{R}^n\}$



so b is reachable exactly when some x reaches it

EXISTENCE $\boxed{Ax = b \text{ is consistent} \iff b \in \mathrm{Col}(A)}$

one sentence the column space controls existence

WORKED EXAMPLE

$\mathrm{Col}(A)$ is $y_3 = 2y_1 + y_2$: $(3, 2, 8)$ passes since $2(3) + 2 = 8$; $(3, 2, 9)$ fails.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.16 range, p. 61. Springer, 2024.

Linear Systems and Solution Sets

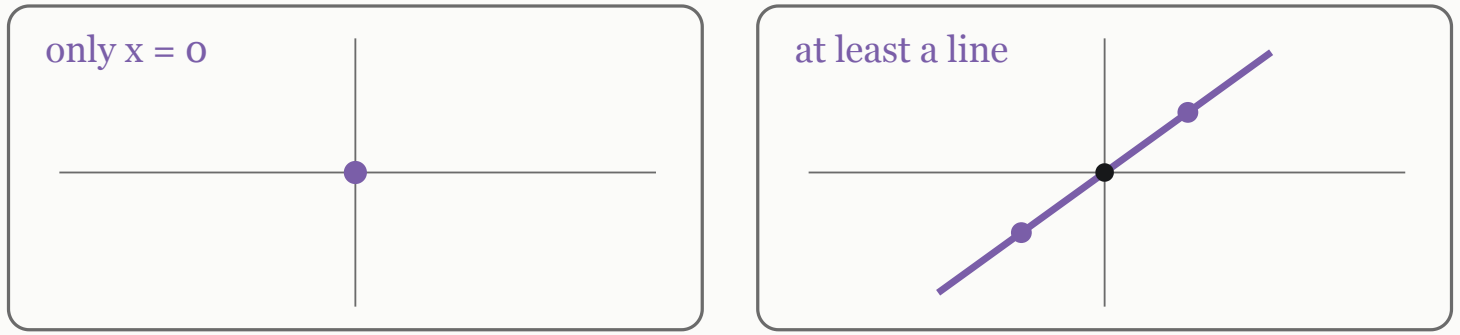
SUPPOSE $Ax_1 = b, \quad Ax_2 = b \implies A(x_1 - x_2) = 0 \implies x_1 - x_2 \in \text{Null}(A)$

so if $\text{Null}(A) = \{0\}$ then $x_1 = x_2$

UNIQUENESS $\boxed{\text{Null}(A) = \{0\} \implies \text{at most one solution}}$

careful at most one, not one

if consistent then it means exactly one

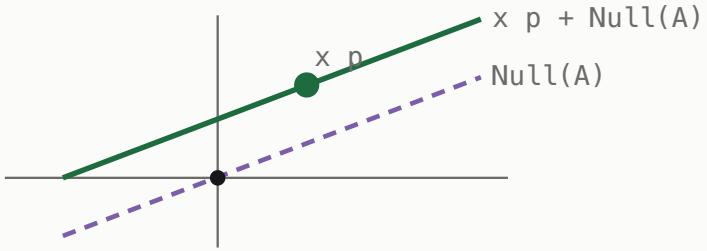


Either only the origin, or at least a whole line of erased inputs.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.15 injectivity, p. 60. Springer, 2024.

Linear Systems and Solution Sets

PARTICULAR $Ax_p = b$ for one particular x_p



ONE SOLUTION, THEN EVERYTHING A IGNORES

`any one` it does not matter which solution you found

GENERAL $\mathcal{S}(A, b) = x_p + \text{Null}(A) = \{x_p + z : z \in \text{Null}(A)\}$

`recipe` find one solution, then add everything A ignores

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Linear Systems and Solution Sets

EACH ONE WORKS

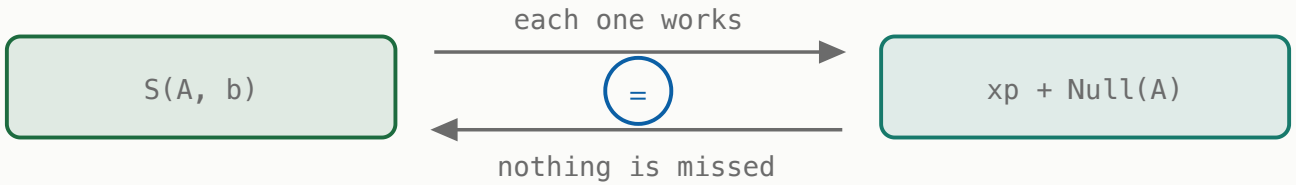
$$A(x_p + z) = b + 0 = b$$

uses linearity and $Az = 0$

NOTHING IS MISSED

$$Ax = b \Rightarrow A(x - x_p) = 0$$

so $x = x_p + z$ with $z = x - x_p \in \text{Null}(A)$



EQUALITY $\boxed{\mathcal{S}(A, b) = x_p + \text{Null}(A)}$

both ways one inclusion each, which is what an equality of sets needs

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Linear Systems and Solution Sets

$$\text{AUGMENT}\quad \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 3 & 8 \end{array}\right] \xrightarrow{r_3 \leftarrow r_3 - 2r_1} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 2 \end{array}\right] \xrightarrow{r_3 \leftarrow r_3 - r_2} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

step 1 $(2,1,3|8) - 2(1,0,1|3) = (0,1,1|2)$

step 2 $(0,1,1|2) - (0,1,1|2) = (0,0,0|0)$

row 3 $0 = 0$: consistent

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Linear Systems and Solution Sets

READ OFF $x_1 + x_3 = 3, \quad x_2 + x_3 = 2, \quad x_3 = t \implies x_1 = 3 - t,\ x_2 = 2 - t$

`free` no pivot in column 3: choose x_3

$t = 0$ one particular solution, $x_p = (3, 2, 0)^{\mathsf{T}}$

ALL OF THEM $\boxed{x = (3, 2, 0)^{\mathsf{T}} + t\,(-1, -1, 1)^{\mathsf{T}}, \qquad t \in \mathbb{R}}$

$t = 1$ $(2, 1, 1)^{\mathsf{T}}$: $2 + 1 = 3$, $1 + 1 = 2$, $4 + 1 + 3 = 8$

$b = (3, 2, 9)^{\mathsf{T}}$ same steps: $9 - 6 = 3$, $3 - 2 = 1$, so row 3 reads $0 = 1$: no solution

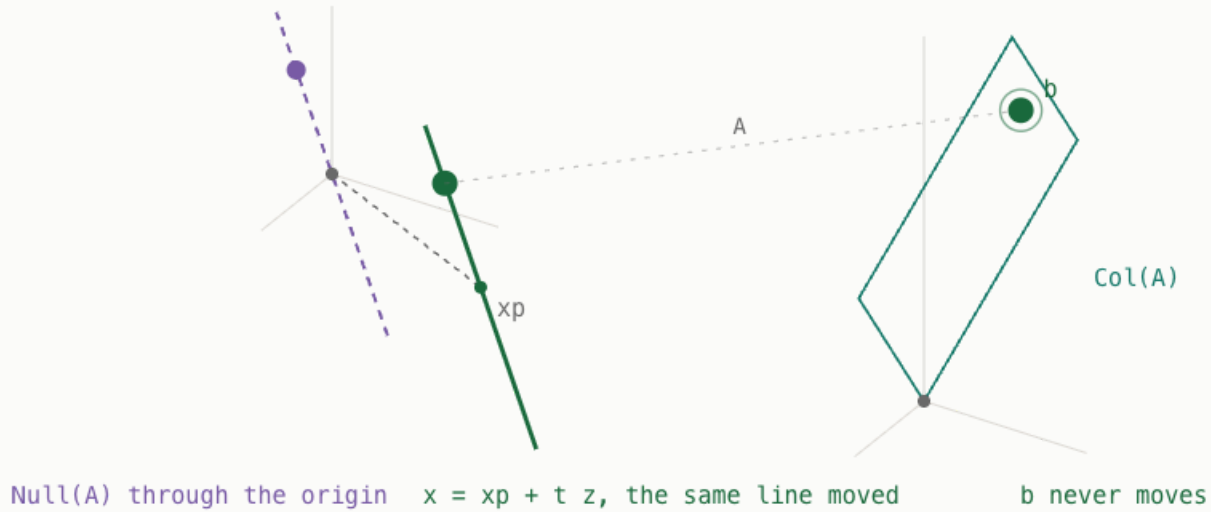
Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.11 null space, p. 59. Springer, 2024.

Linear Systems and Solution Sets

AFFINE SET $\boxed{x_0 + S = \{x_0 + s : s \in S\} \text{ for a subspace } S}$

here $\mathcal{S}(A, b) = x_p + \text{Null}(A)$: the null space, translated

through 0? only if $b = 0$; then $\mathcal{S} = \text{Null}(A)$, a subspace



The null space through the origin, the same line moved to a particular solution, and an output that never moves.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 1.34 conditions for a subspace, p. 18. Springer, 2024.

Linear Systems and Solution Sets

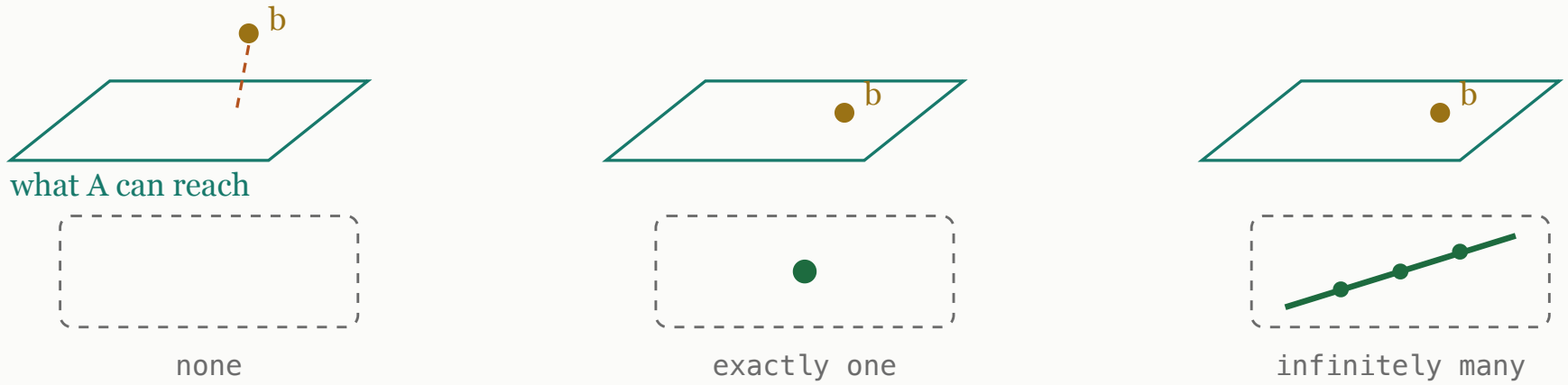
NOTATION $r = \operatorname{rank}(A)$

`unique` when consistent, means $r = n$

CASES

$$
\begin{aligned}
b \notin \operatorname{Col}(A) &: \text{ none} \\
b \in \operatorname{Col}(A),\ \operatorname{Null}(A) = \{0\} &: \text{ one} \\
b \in \operatorname{Col}(A),\ \dim \operatorname{Null}(A) > 0 &: \text{ infinitely many}
\end{aligned}
$$

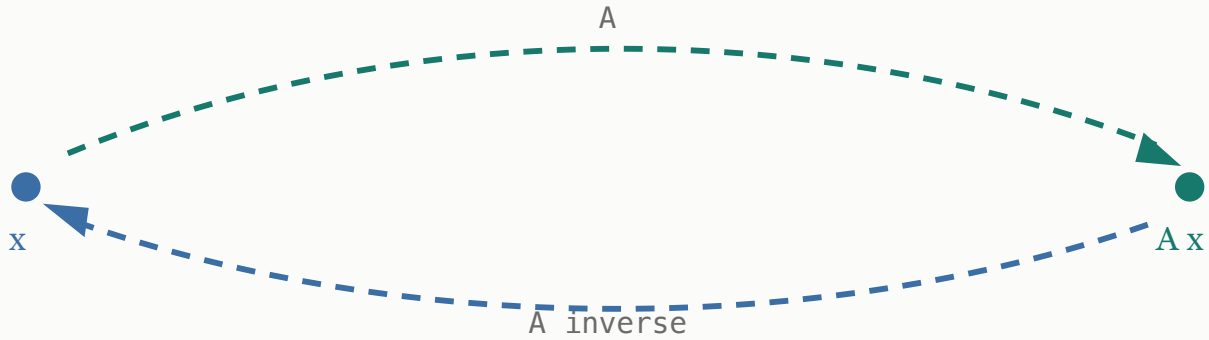
`free` $n - r$ directions



Top: the demand. Bottom: the solutions.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Invertibility and Singular Matrices



Invertibility and Singular Matrices

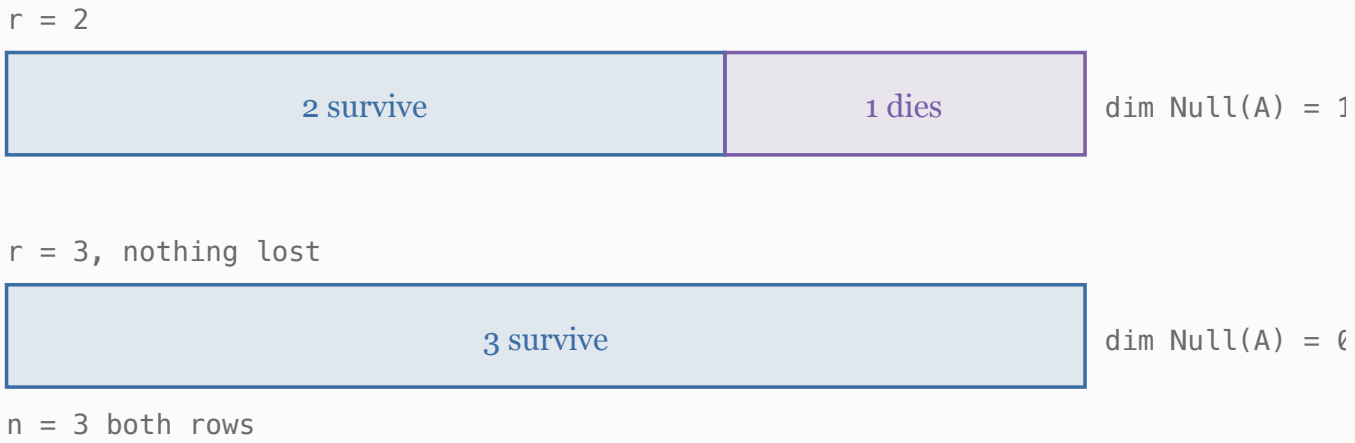
DEMAND $\text{Col}(A) = \mathbb{R}^m$ and $\text{Null}(A) = \{0\}$

`reach` $r = m$: nothing is missed

`no loss` $r = n$ by rank--nullity

`so` $\boxed{m = n \text{ and } r = n}$

`square` a conclusion, not a hypothesis



Both cases have three input directions to spend. Only the top spends one on the null space.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21 fundamental theorem of linear maps, p. 62. Springer, 2024.

Invertibility and Singular Matrices

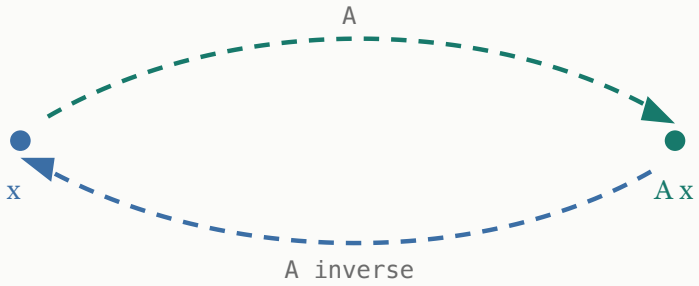
INVERSE $\boxed{AA^{-1} = A^{-1}A = I}$

shapes $A, A^{-1} \in \mathbb{R}^{n\times n}$ and $I = I_n$

both sides two-sided by definition; for square A either half implies the other

SINGULAR $\boxed{A \text{ singular} \iff A \text{ has no inverse}}$

nonsingular the same as invertible



WORKED EXAMPLE

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.79 identity, p. 90; 3.80 invertible and the word singular, p. 91; 3.60 inverse is unique, p. 82. Springer, 2024.

Invertibility and Singular Matrices

IN GENERAL A invertible $\iff$ A injective and surjective

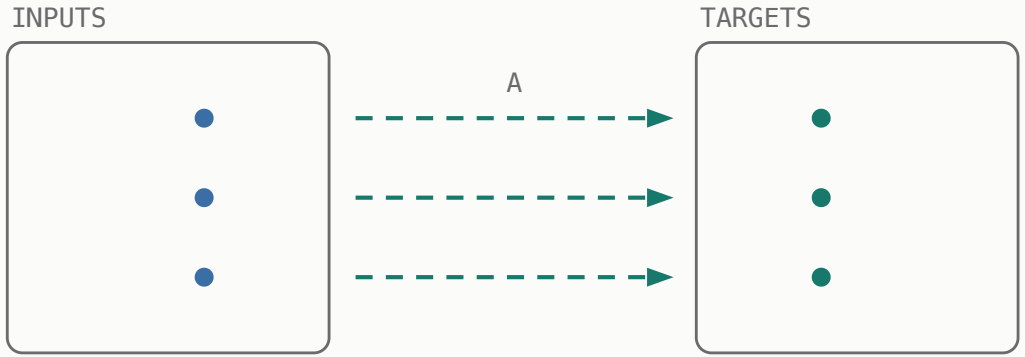
injective $Ax = Ay \Rightarrow x = y$

surjective $\mathrm{Col}(A) = \mathbb{R}^m$

IF A IS SQUARE $\boxed{\mathrm{Null}(A) = \{0\} \iff \mathrm{Col}(A) = \mathbb{R}^n \iff A \text{ invertible}}$

saving check one side, the other follows

why rank--nullity: $r = n$ makes both true at once



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.63 invertibility, p. 83; 3.65 injectivity equivalent to surjectivity, p. 84. Springer, 2024.

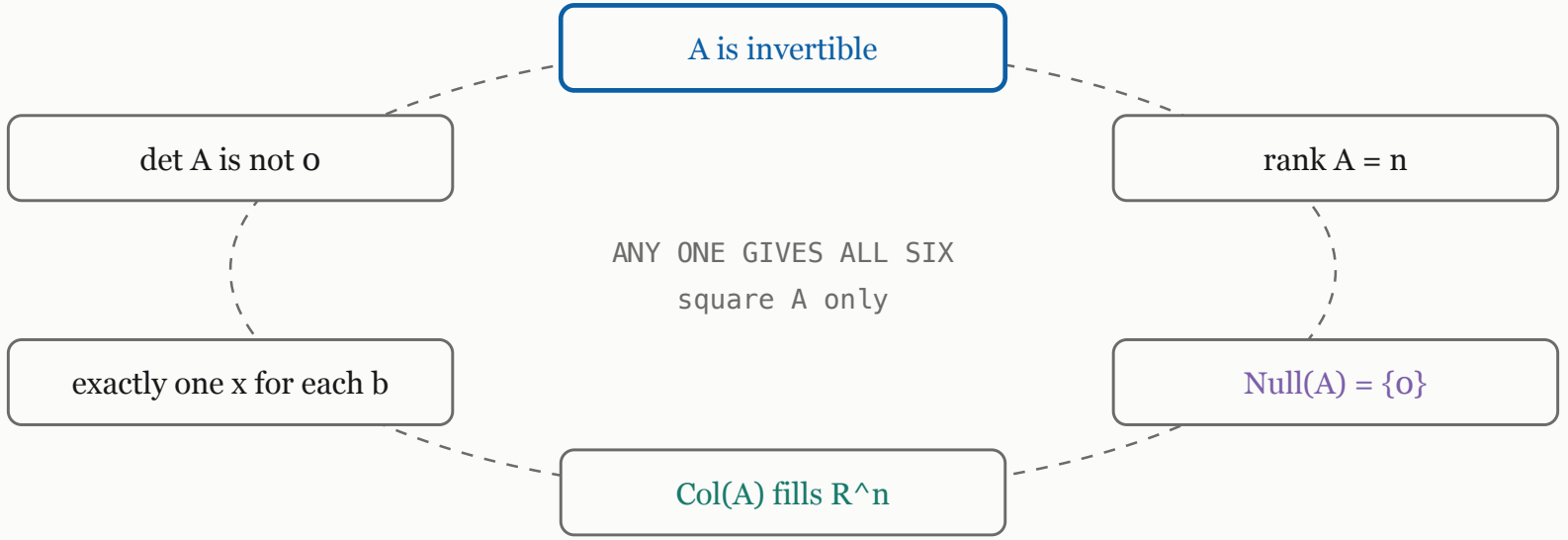
Invertibility and Singular Matrices

SQUARE A

$$A \text{ invertible} \iff r = n \iff \text{Null}(A) = \{0\}$$
$$\iff \text{columns independent} \iff \text{Col}(A) = \mathbb{R}^n$$

and $\iff Ax = b$ has one solution for every b

det a preview; the determinant is a later lecture



Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.63 invertibility, p. 83; 3.65, p. 84; 3.21, p. 62. Springer, 2024.

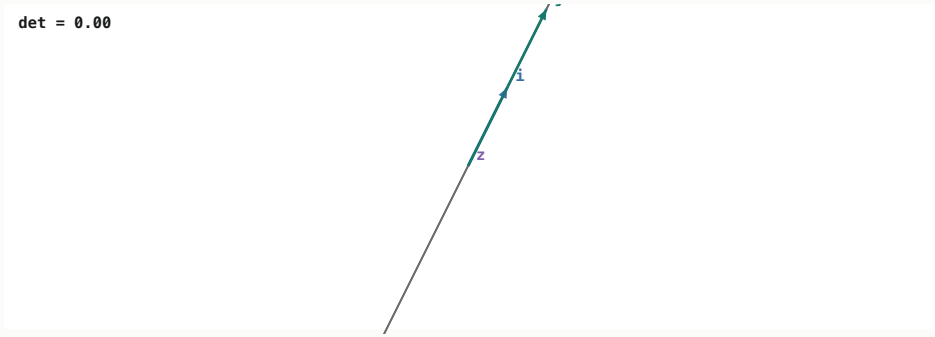
Invertibility and Singular Matrices

SINGULAR $\boxed{\text{Null}(B) \neq \{0\} \iff B \text{ singular}}$

B square and rank deficient

WORKED EXAMPLE

$B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ has $r = 1$, and $B(2, -1)^{\mathsf{T}} = 0$: a nonzero input lands on zero.



The unit square flattens; the area, and the inverse, are gone.

Sanderson, G. ***Essence of Linear Algebra***, ch. 7. 3Blue1Brown, 2016 · Axler, 4th ed., 3.80, p. 91.

Invertibility and Singular Matrices

PREVIEW · LEAST SQUARES

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}, \qquad X^{\mathsf{T}}X = \begin{bmatrix} 1+4+9 & 2+8+18 \\ 2+8+18 & 4+16+36 \end{bmatrix} = \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix}$$

the fit $\min_{\beta} \|X\beta - y\|^2 \;\Rightarrow\; X^{\mathsf{T}}X\beta = X^{\mathsf{T}}y$

entries $(X^{\mathsf{T}}X)_{ij}$ is column i dotted with column j; the off-diagonal pair agree

Goodfellow, I., Bengio, Y. and Courville, A. ***Deep Learning***, ch. 2. MIT Press, 2016.

Invertibility and Singular Matrices

SINGULAR $$X\begin{bmatrix}2\\-1\end{bmatrix} = \begin{bmatrix}2-2\\4-4\\6-6\end{bmatrix} = 0 \implies X^{\mathsf{T}}X\begin{bmatrix}2\\-1\end{bmatrix} = 0$$

collinear column 2 is twice column 1: $(2,-1)^{\mathsf{T}} \in \mathrm{Null}(X)$

NULL SPACES $$\boxed{X^{\mathsf{T}}Xz = 0 \;\Rightarrow\; \|Xz\|^2 = z^{\mathsf{T}}X^{\mathsf{T}}Xz = 0 \;\Rightarrow\; Xz = 0}$$

so $\mathrm{Null}(X^{\mathsf{T}}X) = \mathrm{Null}(X)$

the fit $\beta + t(2,-1)^{\mathsf{T}}$ keeps $X\beta$: not identifiable

Goodfellow, I., Bengio, Y. and Courville, A. ***Deep Learning***, ch. 2. MIT Press, 2016.

Invertibility and Singular Matrices

AT A SINGULARITY $J(q)$ loses rank $\Longrightarrow$ a tool direction is lost

rank drop $\mathrm{Col}(J)$ shrinks: some $\dot{x}$ has no $\dot{q}$

nearby joint rates grow without bound for tool directions along the lost one

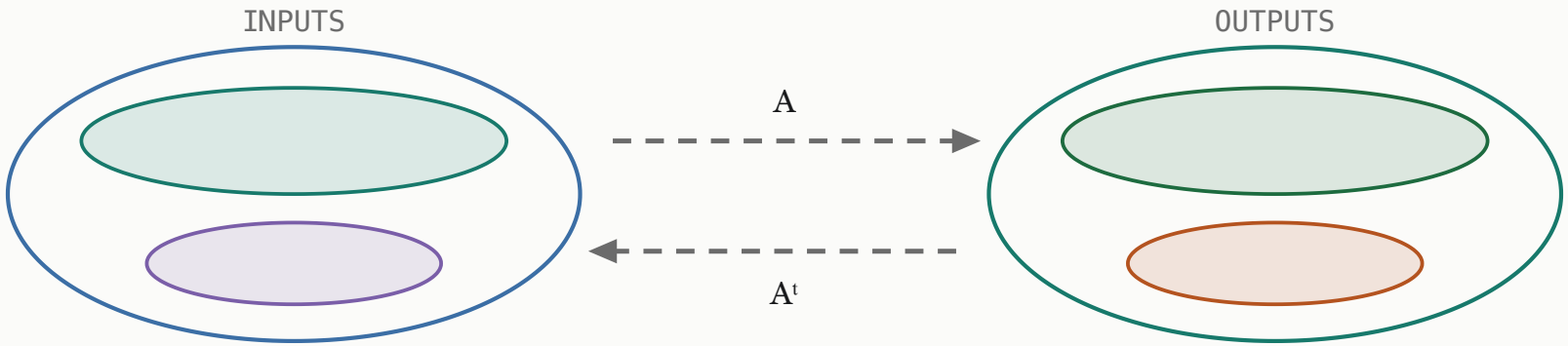
NEXT $\mathrm{Col}(A),\ \mathrm{Null}(A),\ \mathrm{Col}(A^{\mathsf{T}}),\ \mathrm{Null}(A^{\mathsf{T}})$

one matrix four spaces, one rank



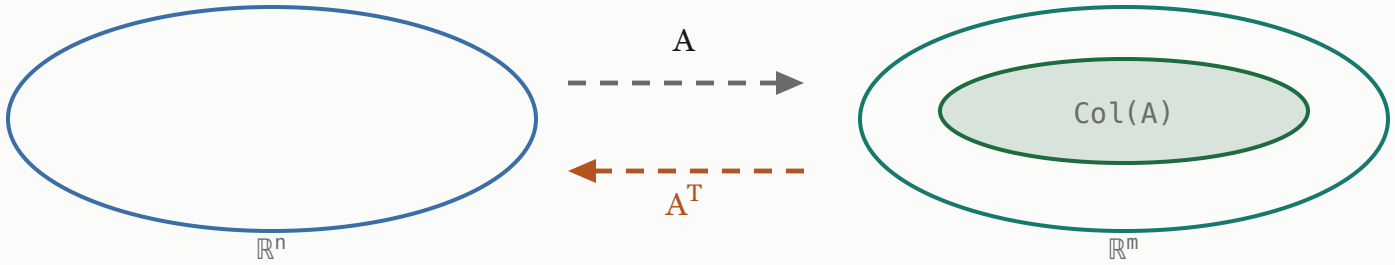
Lynch, K. M. and Park, F. C. ***Modern Robotics***, §5.3 singularity analysis, p. 191. Cambridge University Press, 2017 preprint.

The Four Fundamental Subspaces



The Four Fundamental Subspaces

SO FAR $A : \mathbb{R}^n \to \mathbb{R}^m$, $\quad \mathrm{Col}(A)$ and $\mathrm{Null}(A)$



but $A^\mathsf{T} : \mathbb{R}^m \to \mathbb{R}^n$ is a linear map too

SO ASK **what spaces come from A^T?**

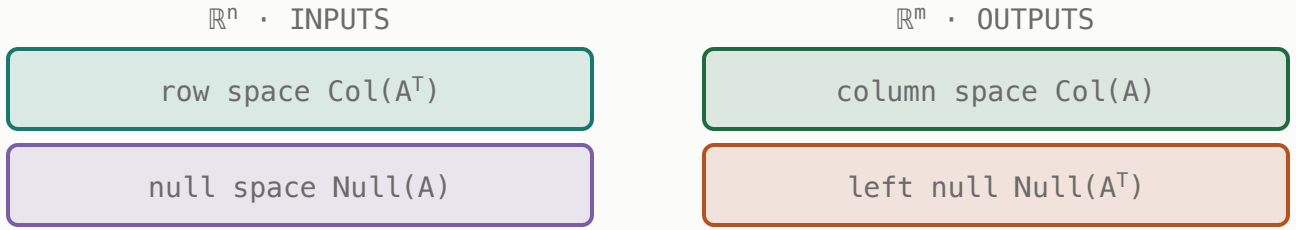
the answer $\mathrm{Col}(A^\mathsf{T})$ and $\mathrm{Null}(A^\mathsf{T})$: two more, for free

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.54 transpose, p. 77. Springer, 2024.

The Four Fundamental Subspaces

NEW NAMES

$$\mathrm{Row}(A) = \mathrm{Col}(A^\mathsf{T}), \qquad \mathrm{Null}(A^\mathsf{T}) = \{y : A^\mathsf{T} y = 0\}$$



left null $A^\mathsf{T} y = 0$ is $y^\mathsf{T} A = 0^\mathsf{T}$: y multiplies from the left

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.54 transpose, p. 77; 3.11 null space, p. 59. Springer, 2024.

The Four Fundamental Subspaces

OUTPUT SIDE

$$\mathrm{Col}(A) = \mathrm{span}\{(1,0,2)^{\mathsf{T}}, (0,1,1)^{\mathsf{T}}\}$$

$$\mathrm{Null}(A^{\mathsf{T}}) = \mathrm{span}\{(-2,-1,1)^{\mathsf{T}}\}$$

INPUT SIDE

$$\mathrm{Row}(A) = \mathrm{span}\{(1,0,1)^{\mathsf{T}}, (0,1,1)^{\mathsf{T}}\}$$

$$\mathrm{Null}(A) = \mathrm{span}\{(-1,-1,1)^{\mathsf{T}}\}$$

```
row 1   1 0 1
row 2   0 1 1      2r1 + r2 = r3
row 3   2 1 3      so −2r1 − r2 + r3 = 0
```

so $\boxed{y = (-2,-1,1)^{\mathsf{T}}, \qquad A^{\mathsf{T}}y = 0}$

`read it` the row weights

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.57 column rank equals row rank, p. 78. Springer, 2024.

The Four Fundamental Subspaces

RANK EQUALITY $$\boxed{\operatorname{rank}(A) = \operatorname{rank}(A^{\mathsf{T}}) = r}$$



the output side: r = 2, m − r = 1

so column rank equals row rank

ALL FOUR $$\boxed{\begin{aligned} \dim \operatorname{Col}(A) &= r & \dim \operatorname{Null}(A) &= n - r \\ \dim \operatorname{Col}(A^{\mathsf{T}}) &= r & \dim \operatorname{Null}(A^{\mathsf{T}}) &= m - r \end{aligned}}$$

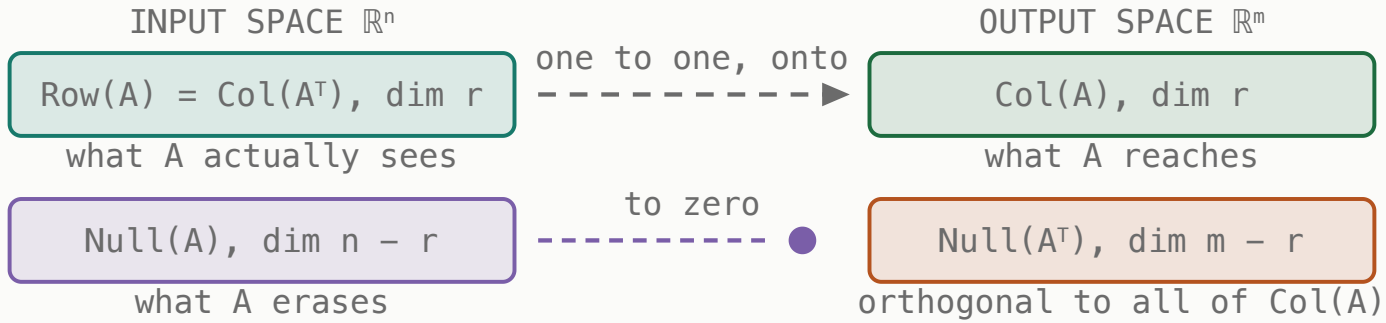
two sums $r + (n - r) = n$ and $r + (m - r) = m$

WORKED EXAMPLE

Our A: $r = 2$, so $2 + 1 = 3$ columns and $2 + 1 = 3$ rows.

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.57 column rank equals row rank, p. 78; 3.21, p. 62. Springer, 2024.

The Four Fundamental Subspaces



LECTURE 1 IN ONE LINE $\dim = r,\ n-r,\ r,\ m-r$

later the two pairs form orthogonal-complement decompositions

Axler, S. ***Linear Algebra Done Right***, 4th ed., 3.21, p. 62; 3.57, p. 78. Springer, 2024.

INTEGRATED WHITEBOARD PROBLEM

Linear Systems and Four Subspaces

Given $C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 3 & 1 \end{bmatrix}$, $b = (4, 1, 5)^{\mathsf{T}}$. Find reachability, all solutions and four-space dimensions.

$$r_3 - r_1 - r_2 = 0 \;\Rightarrow\; b_3 = b_1 + b_2, \quad r = 2$$

`r` rank; two independent rows

$$x = (2, 1, 0)^{\mathsf{T}} + t(2, -1, 1)^{\mathsf{T}}$$

`t` any real; dimensions: $2, 1, 2, 1$

YOUR TURN · NEW MATRIX

Linear Systems and Four Subspaces

$D = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$, $b = (3, 1)^{\mathsf{T}}$. Find all solutions. Is every target reachable? Is its input unique?

$$x_3 = t, \quad x_1 = 3 - 2t, \quad x_2 = 1 + t$$

`t` the free third coordinate

$$x = (3, 1, 0)^{\mathsf{T}} + t(-2, 1, 1)^{\mathsf{T}}$$

`rank 2` every target reached; nullity 1: inputs not unique

Courses, books and lectures

1. Strang, G. *18.06 Linear Algebra*, Lectures 3, 7 and 10. MIT OCW, 2010. CC BY-NC-SA 4.0.
2. Grizzle, J. W. *ROB 501: Mathematics for Robotics*, Michigan. CC BY-NC 4.0.
3. Axler, S. *Linear Algebra Done Right*, 4th ed., 2024. Open access.
4. Goodfellow, Bengio and Courville. *Deep Learning*, MIT Press, 2016, ch. 2.
5. Sanderson, G. *Essence of Linear Algebra*, ch. 2, 3 and 7. 3Blue1Brown, 2016.

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Papers and books

6. Strang, G. The fundamental theorem of linear algebra. *Amer. Math. Monthly*, 1993.
7. Black, K. et al. π0. *RSS*, 2025; arXiv:2410.24164, posted 2024.
8. Lipman, Y. et al. Flow Matching for Generative Modeling. *ICLR*, 2023. arXiv:2210.02747.
9. Lynch and Park. *Modern Robotics*, §5.3, §6.3, 2017.
10. Barfoot, T. D. *State Estimation for Robotics*, ch. 4. Cambridge, 2017.